AY: 2025-2026
Course: PPOM 1
Class of 2029
Week 13, Lecture 111

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Orthopedics: Lower Extremity

Presented by:

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Learning Objectives:

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1. To recognize bony and soft tissue injuries to the lower extremities
2. To appreciate complications related to lower extremities
3. To understand management options in lower extremities injuries
4. To recognize bony and soft tissue injuries to the lower extremities
5. To identify urgent and emergent lower extremities conditions
6. To interpret and apply clinical signs and symptoms of lower extremities processes in the differential diagnosis.

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RED WOLVES and BEARS

Hip

A. Anatomy
B. Sports Injuries
C. Hip Dislocations
D. Hip Fractures
E. Hip (and knee) degeneration + replacement

Hip Anatomy

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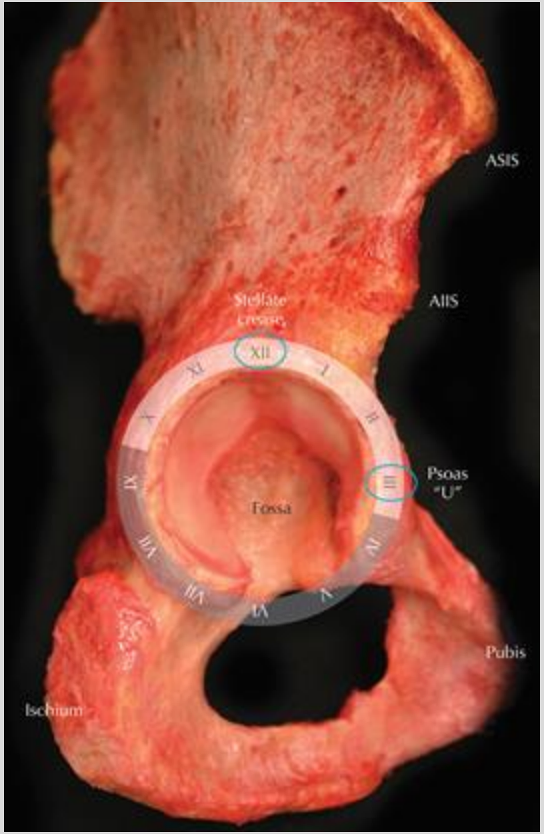

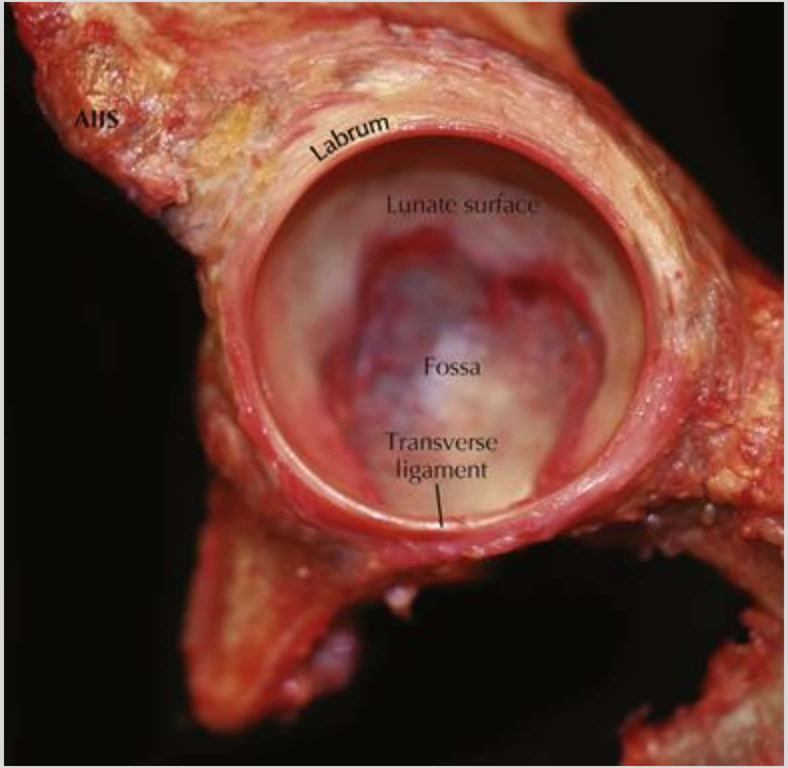

Madden, Christopher. Netter's Sports Medicine. 2023

Hip Anatomy

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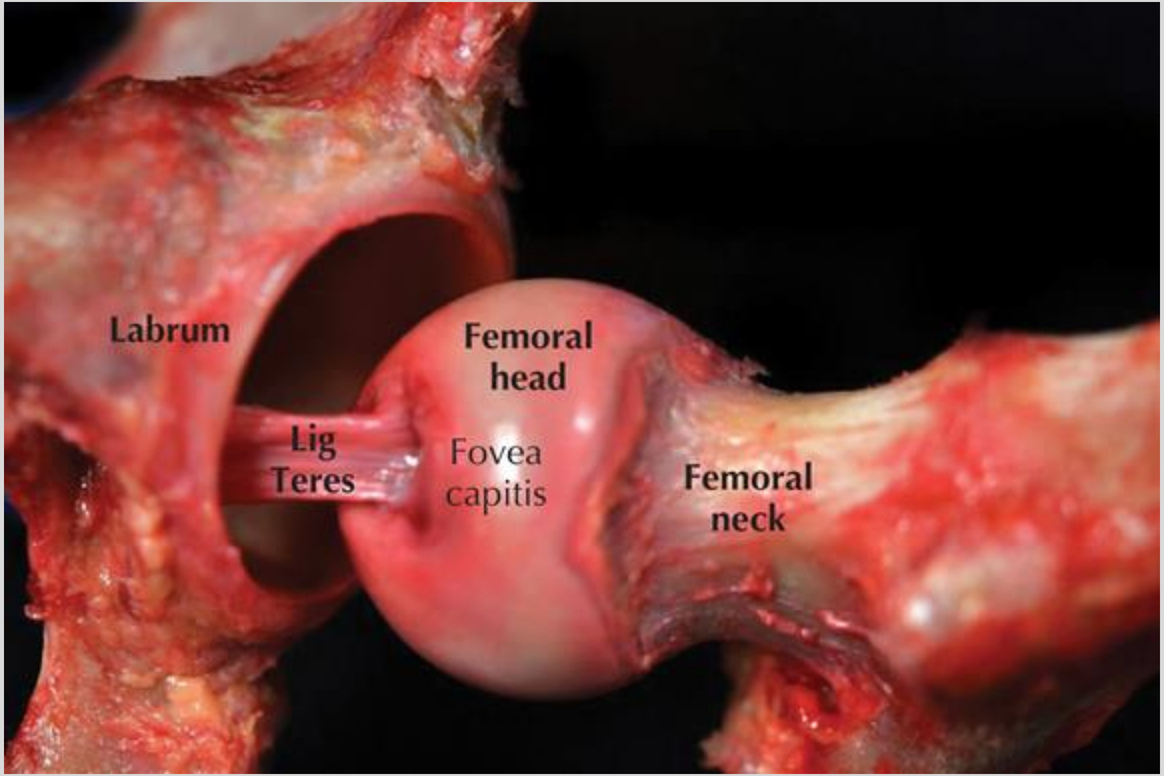

Madden, Christopher. Netter's Sports Medicine. 2023

Pelvic Girdle

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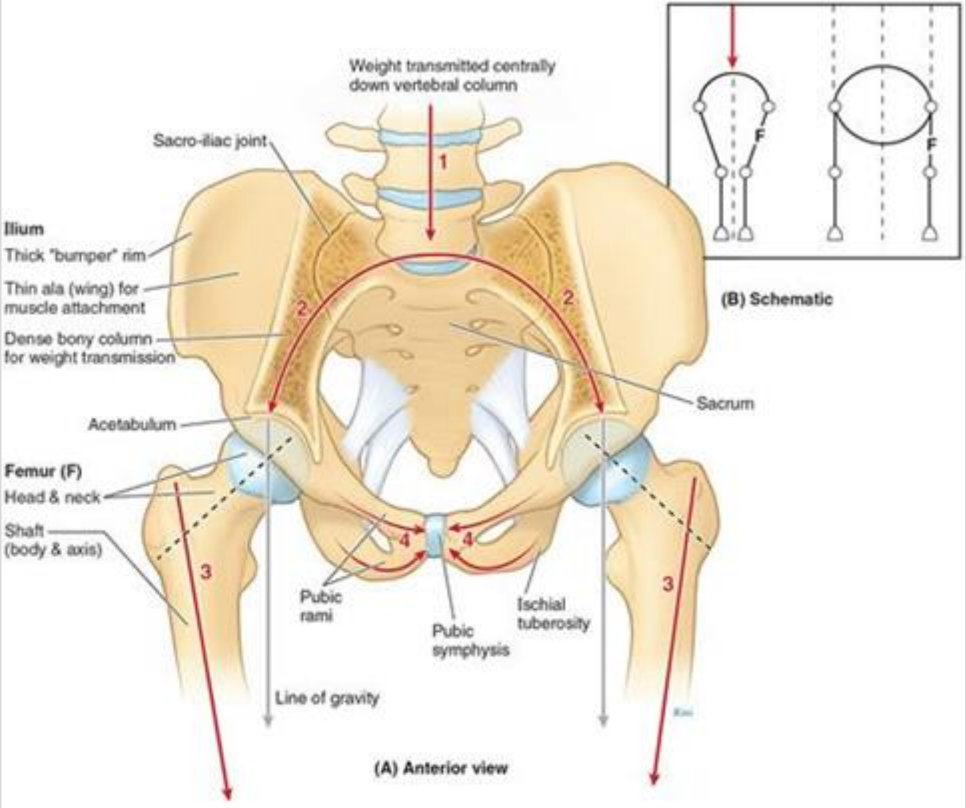
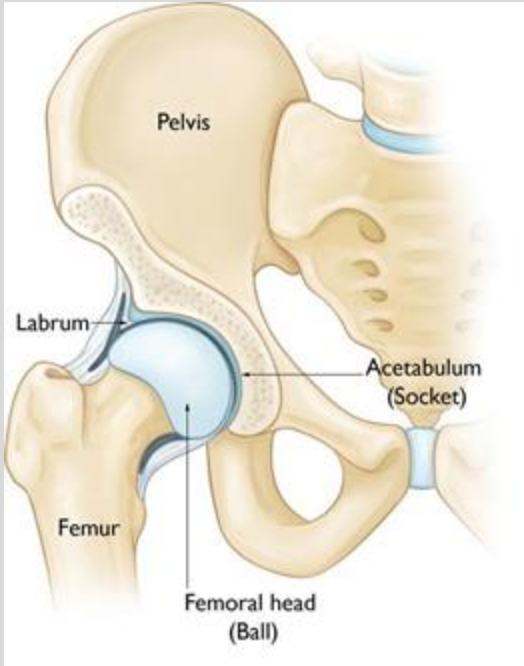
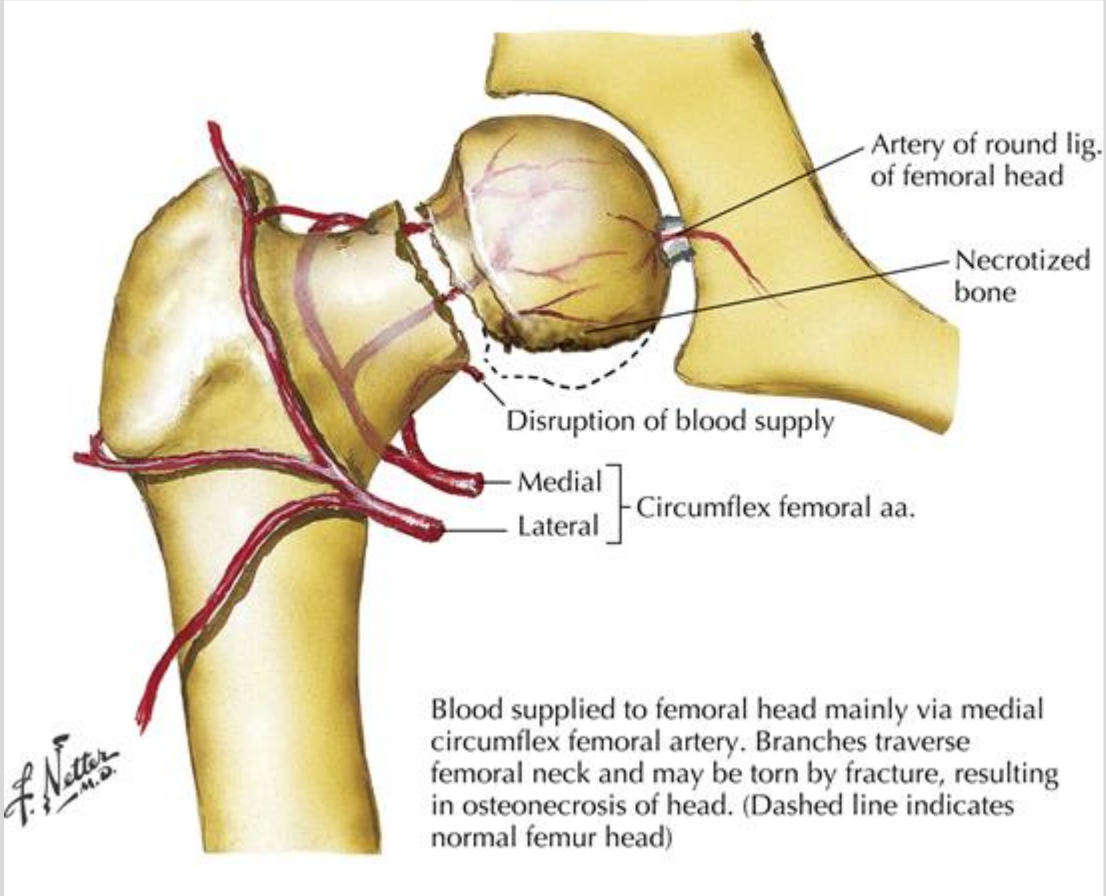




https://orthoinfo.aaos.org

- Moore's Clinically Oriented Anatomy, 9e, Ch. 7

Blood Supply to the Hip



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Mainly the medial circumflex femoral artery.

Intracapsular branches are the retinacular arteries.

- Cleland, Joshua A. Netter's Orthopaedic Clinical Examination. 4th Edition. 2022.

Sciatic Nerve

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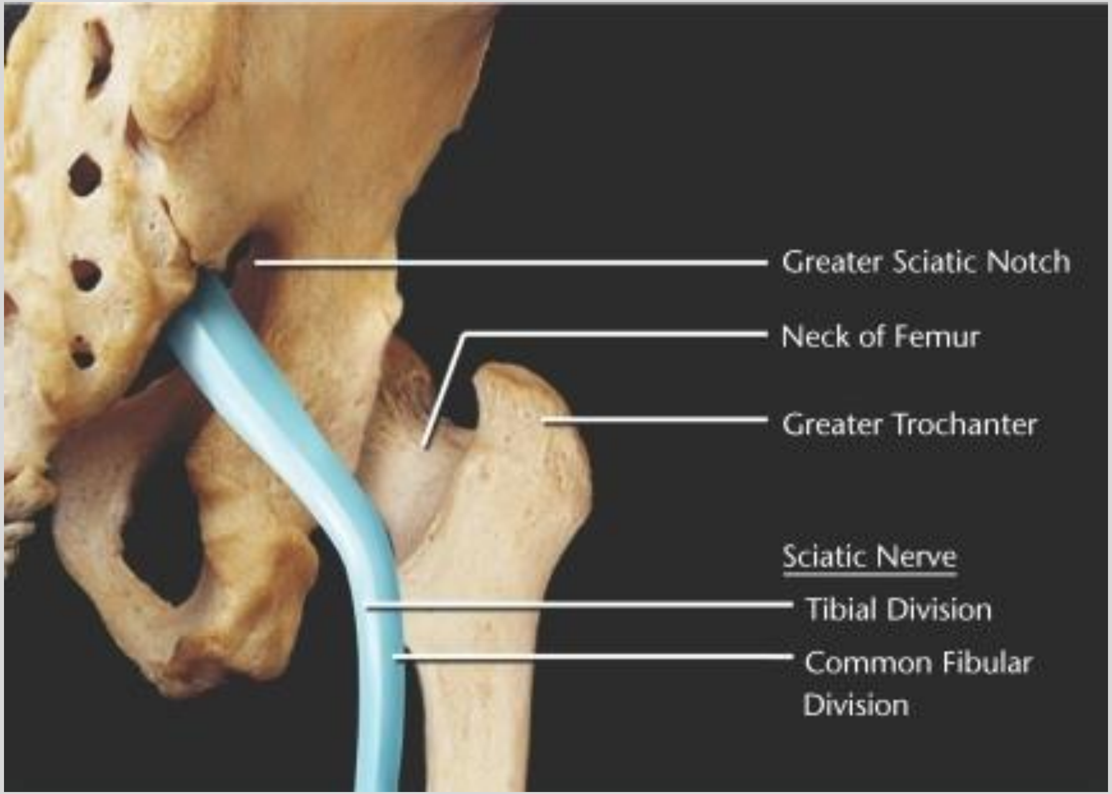

Yates C, Bandy WD, Blasier RD. Traumatic dislocation of the hip in a high school football player. *Phys Ther*. 2008;88(6):780-788. doi:10.2522/ptj.20070298

Proximal Hamstring Injuries

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a. Typically occurs during sprinting when there is a sudden stretch on the musculotendinous junction
b. Severe → Complete avulsion may occur at the ischium
c. Symptoms→ Sudden pain during exercising at the posterior thigh up to the ischium
d. May also report some numbness and tingling in the area corresponding to the sciatic nerve due to presence of hematoma that compresses the sciatic nerve
e. Treatment
    i. Strain → Supportive (RICE)
    ii. Complete Avulsion Injuries → If the hamstring is retracted more than 3 cm, early surgical intervention. If less than 3 cm of retraction, may try nonoperative treatment

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Rectus Femoris Strains- “Hip Flexor Strain”

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a. More common than complete avulsion
b. Rectus Femoris is m/c strained muscle in the quadriceps group
c. MOI: Attempt to kick a ball (football or soccer) - Higher risk when patient’s foot hits another player in the middle of the kicking motion
d. Symptoms → Acute pain in the midline of the anterior thigh
e. Pain or weakness with resisted hip flexion or knee extension
f. Treatment →
    i. Strain → Supportive (RICE) and rehab
    ii. Proximal rupture → nonoperative vs. operative

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Femoroacetabular Impingement

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a. Results from abnormal contact between the femoral head and the acetabulum
    i. CAM (Femoral head is not round and cannot rotate smoothly inside the acetabulum)
    ii. Pincer (An acetabular abnormality whereby extra bone extends beyond the normal rim of the acetabulum, impinging upon the femoral head and the labrum)
    iii. Mixed (Combination)
b. Symptoms → Limited ROM with groin/hip pain especially with flexion and IR
c. Nonsurgical Treatments → RICE, Physical Therapy, Activity modification
d. Surgical
    i. Acetabular rim trimming
    ii. Labral repair, if needed
    iii. Labral Reconstruction, if needed
    iv. Osteochondroplasty

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Femoroacetabular Impingement

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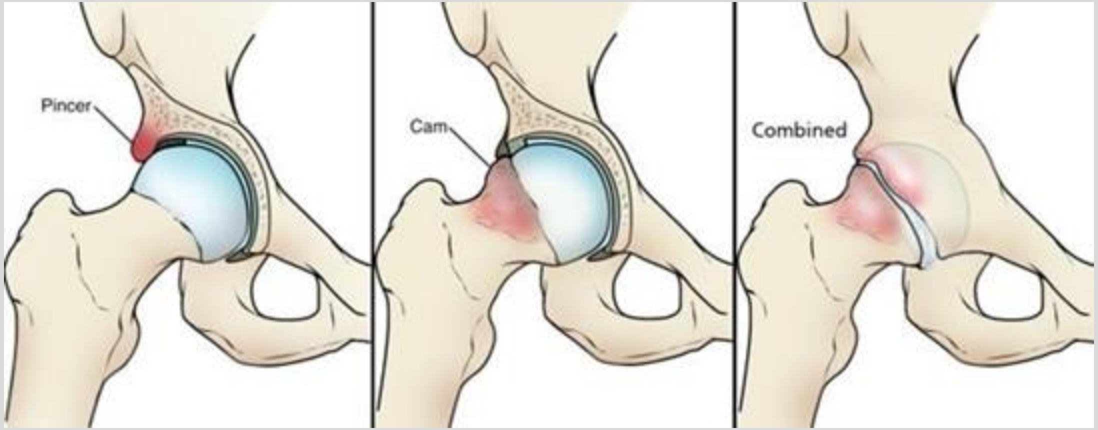

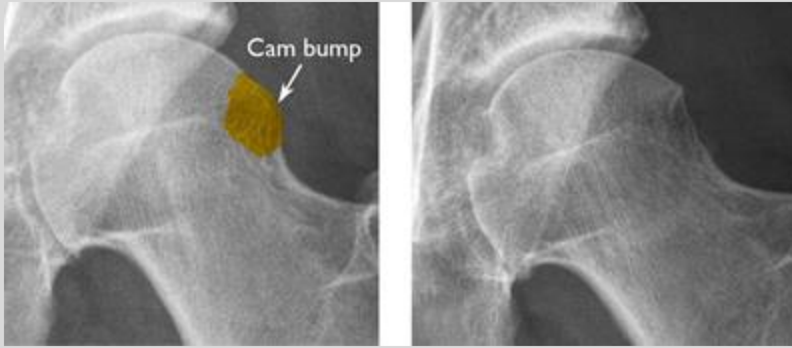

https://orthoinfo.aaos.org/en/diseases--conditions/femoroacetabular-impingement/

Reproduced from Diaz-Ledezma C, Higuera CA, Parvizi J: Mini-open approach for the treatment of FAI in Sierra RJ, ed: Femoracetabular Impingement. Rosemont, IL, American Acad of Orthopaedic Surgeons, 2013, pp 81-91.

Hip Pointer (Iliac Crest Contusion)

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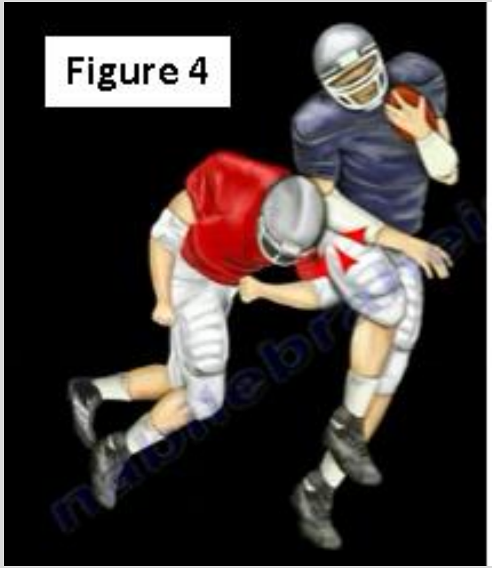
- Contusion of the iliac crest and/or the greater trochanteric region of the femur following a direct impact or collision
- Primary Etiology is a direct blow to the iliac crest or greater trochanter during contact sports
- Most commonly, patients present with varying degrees of bruising around the area of impact

- Imaging may be needed if concerned for fractures

- Treatment → RICE, Anti-inflammatory pain medications.
- As pain improved, initiation of ROM and Active resistance exercises should begin
  - Consider padding at the injury site



Images from: https://www.huffpost.com/entry/hip-pointer_b_13955542

https://www.orthobullets.com/knee-and-sports/3090/iliac-crest-contusion-hip-pointer

https://www.ncbi.nlm.nih.gov/books/NBK538258/

Hip Pointer

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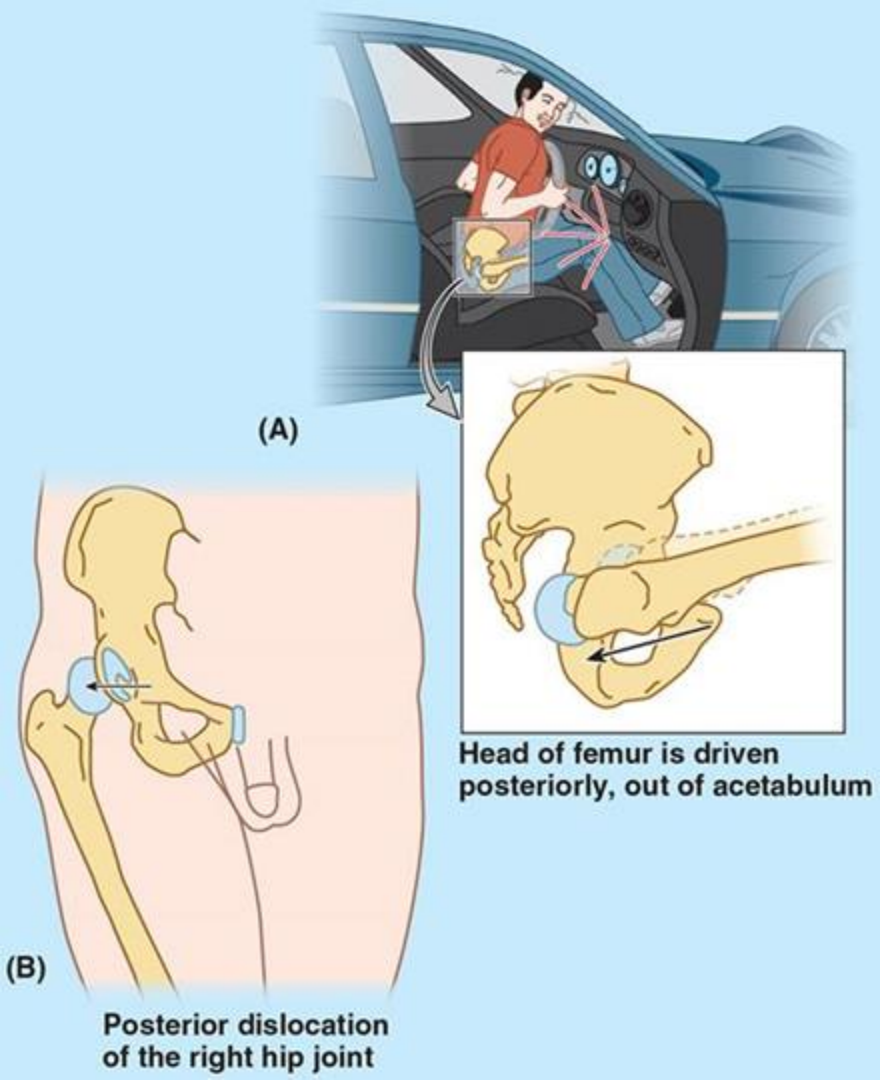
Chargers receiver Keenan Allen attempts to catch a pass Thursday in Kansas City.(LARRY W. SMITH / EFE)

https://www.sandiegouniontribune.com/sports/profootballdoc/sd-sp-pfd-keenan-allen-chargers-hip-injury-1213-story.html

Posterior Hip Dislocations

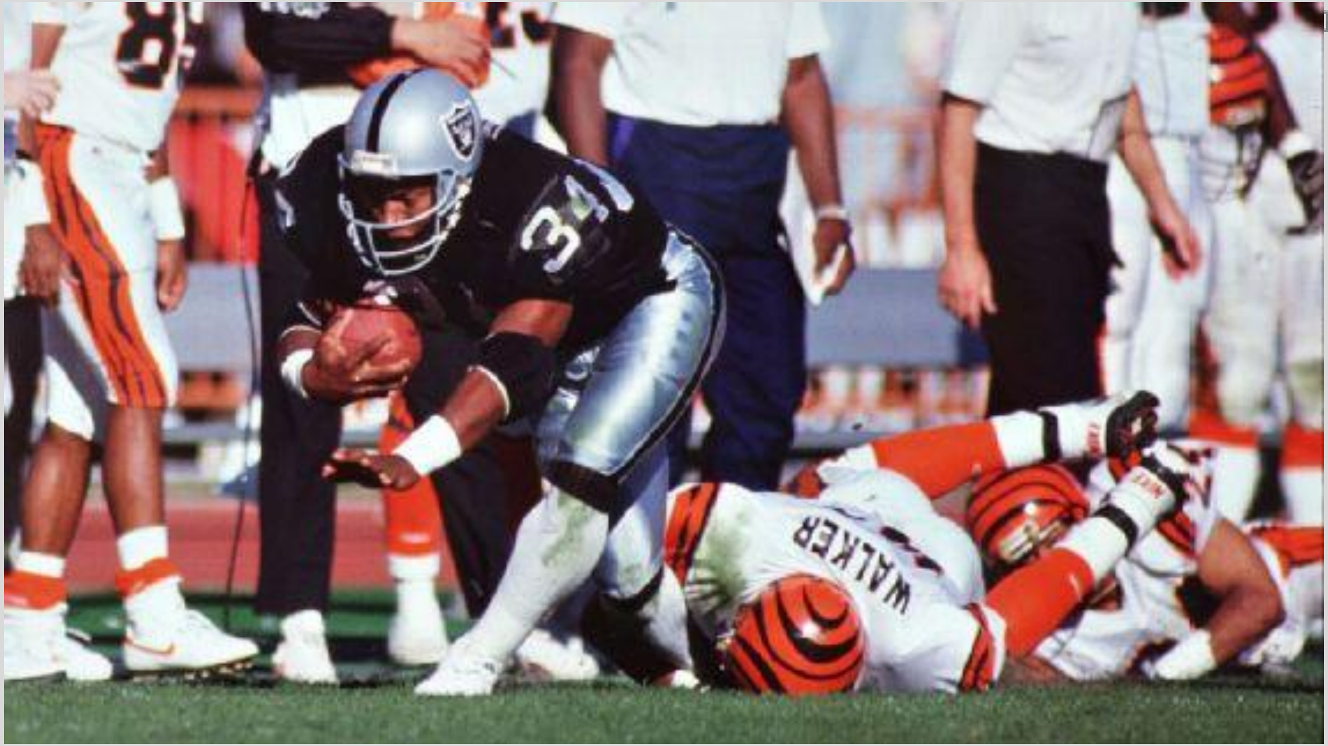
- Most Common dislocation
- Mechanism:
  - High-Energy trauma
  - "Dashboard injury" (force directed posteriorly, commonly on a flexed hip)
- Physical Examination
  - Shortening
  - Flexion, Internal Rotation & Adduction
  - Careful neurologic evaluation

https://usmlepathslides.tumblr.com/post/64398003332/posterior-hip-dislocation-posterior-hip



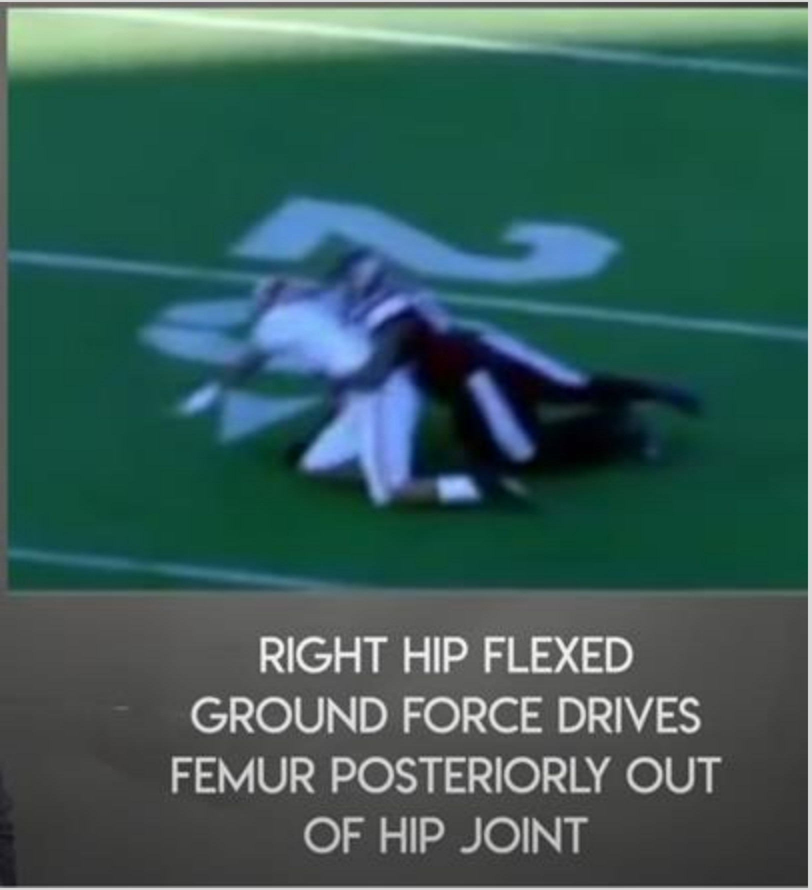
https://radiopaedia.org/articles/posterior-dislocation-of-the-hip?lang=us

Moore's Clinically Oriented Anatomy, 9e, Ch. 7



Bo Jackson suffered a dislocated hip on this play in a playoff game on Jan. 13, 1991. Getty Images

Tua Tagovailoa



Screenshot taken from:
https://www.youtube.com/watch?v=z3QSMsRWXLQ

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Tua Tagovailoa's leg hangs internally rotated with dislocated hip.

https://www.arlingtoncardinal.com/2019/11/alabamas-tua-tagovailoa-assisted-by-walk-off-the-field-when-he-should-have-been-immobilized-and-carried-off-on-a-stretcher/

Anterior Hip Dislocations

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- Less common (10-15% Overall)
- History
- Physical Examination
  - Limited ROM
  - External Rotation, Flexion, Abduction
  - Careful Neurologic Evaluation

Moore’s Clinically Oriented Anatomy, 9e, Ch. 7

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Posterior Dislocation

Anteroposterior view. Dislocated femoral head lies posterior and superior to acetabulum. Femur adducted and internally rotated; hip flexed. Sciatic nerve may be stretched

Anteroposterior radiograph shows posterior dislocation

Allis maneuver. Patient supine on table, under anesthesia or sedation. Examiner applies firm distal traction at flexed knee to pull head into acetabulum; slight rotary motion may also help. Assistant fixes pelvis by pressing on anterior superior iliac spines

Anterior Dislocation

Anterior view. Femoral head in obturator foramen of pelvis; hip flexed and femur widely abducted and externally rotated

Characteristic position of affected limb. Hip flexed, thigh abducted and externally rotated.

- Thompson, Jon C. Netter's Concise Orthopedic Anatomy, 2nd Edition. 2010

Hip Dislocations

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1. Treatments
   a. Early reduction essential (< 6 hours)
      i. Closed reduction (without surgery, just manipulating the bone, trying to “pop” it back in)
      ii. Open reduction (surgically putting the bone in place) if irreducible or if fracture found on x-ray
   b. Repeat x-rays and Neuro exams

1. Complications
   a. Avascular necrosis
   b. Sciatic nerve injury (Posterior dislocations)
   c. Femoral artery/nerve injury (Anterior dislocations)
   d. Osteoarthritis

- Thompson, Jon C. Netter’s Concise Orthopedic Anatomy, 2nd Edition. 2010

Hip Fractures

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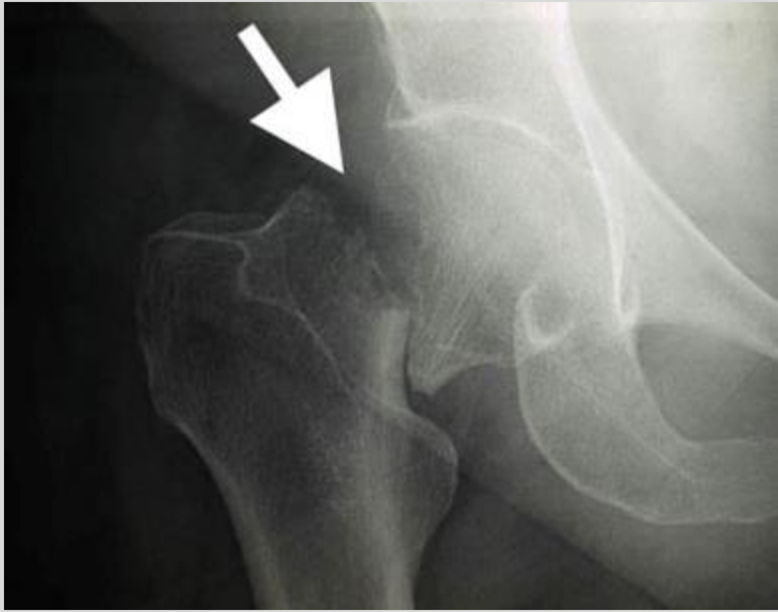
- History and Physical Examination
  - Young Patients
    - Major Trauma
    - Underlying Disease
  - Older Patients (Low Energy)
    - Fall From Standing Position
    - Osteoporotic (weak) bone
- Physical Examination
- Neurovascular Examination

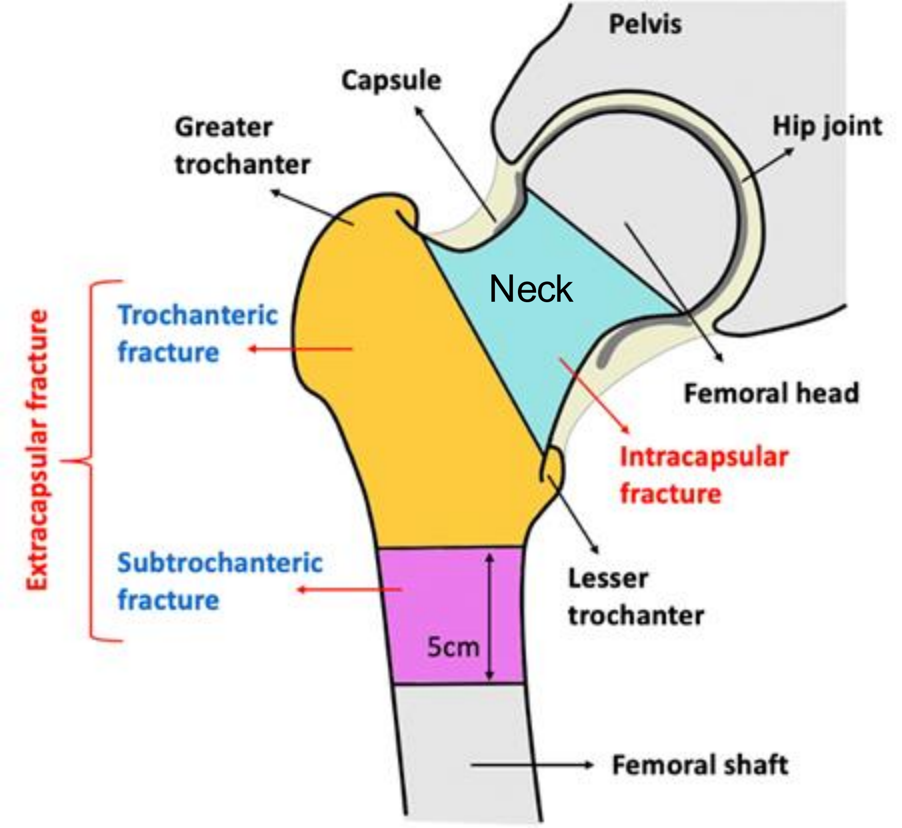


https://orthoinfo.aaos.org/en/diseases--conditions/hip-fractures

Types of Hip Fractures and Vascular Significance

****DON'T MEMORIZE SPECIFIC HIP FRACTURE TYPES****

- Femoral Head and Femoral Neck Fractures
    - Intracapsular Injury
    - Risk of vascular complication
    - Destroys retinacular arteries (branches of the circumflex femoral arteries)
- Intertrochanteric and Subtrochanteric Fractures
    - Extracapsular Injury
    - Less incidence of avascular necrosis because of abundant blood supply.



https://www.researchgate.net/figure/Types-of-hip-fractures_fig4_340718618

Treatment

- Goals:
  - Protect from additional damage
  - Minimize discomfort
  - Restore hip function
  - Allow rapid mobilization
  - Avoid avascular necrosis
- Non-operative
  - Fixation almost always indicated
- Operative Management - depends on type of fracture and patient
  - Pins
  - Screws
  - Hemiarthroplasty
  - Intramedullary Nail

DO NOT MEMORIZE (not tested, just interesting)

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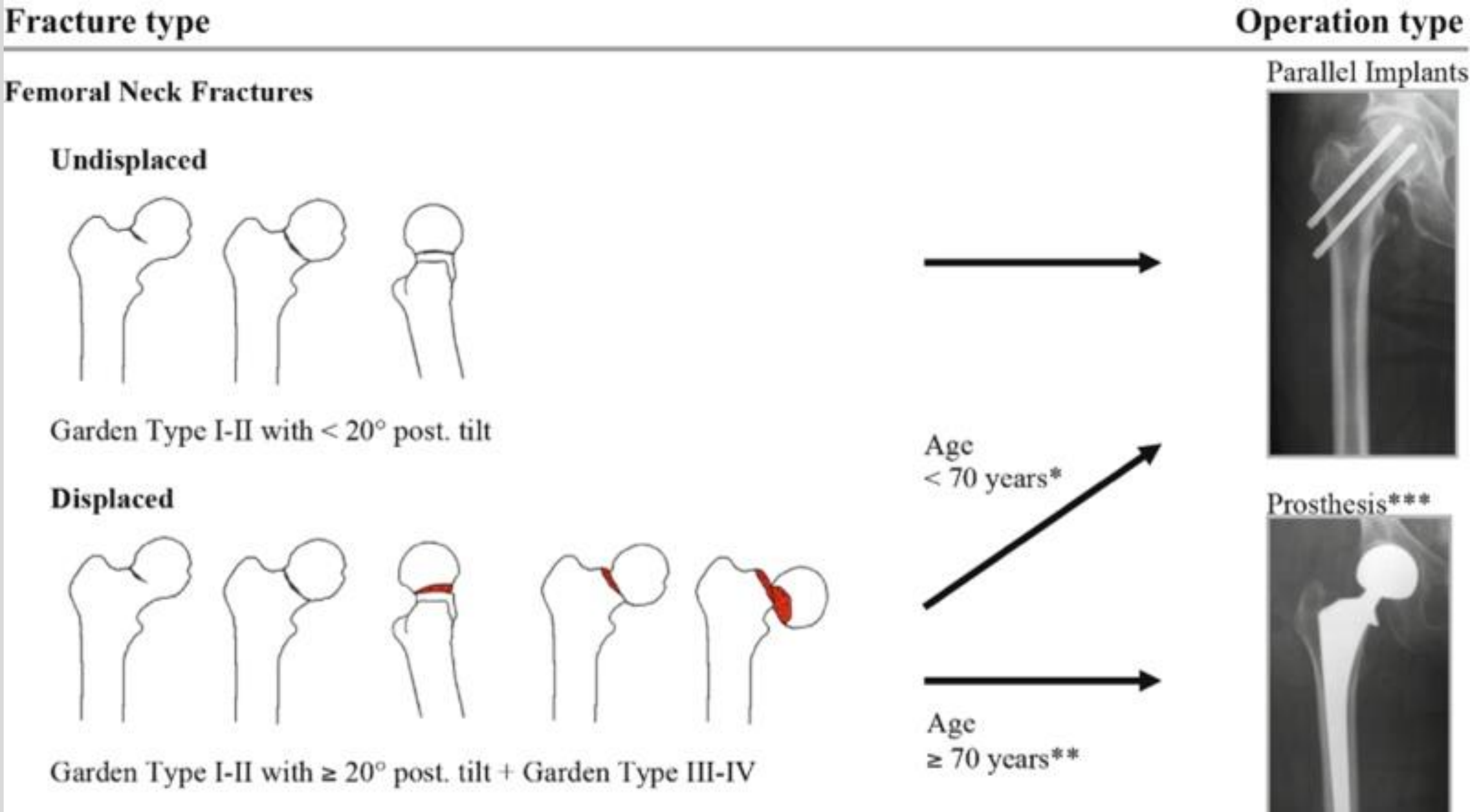
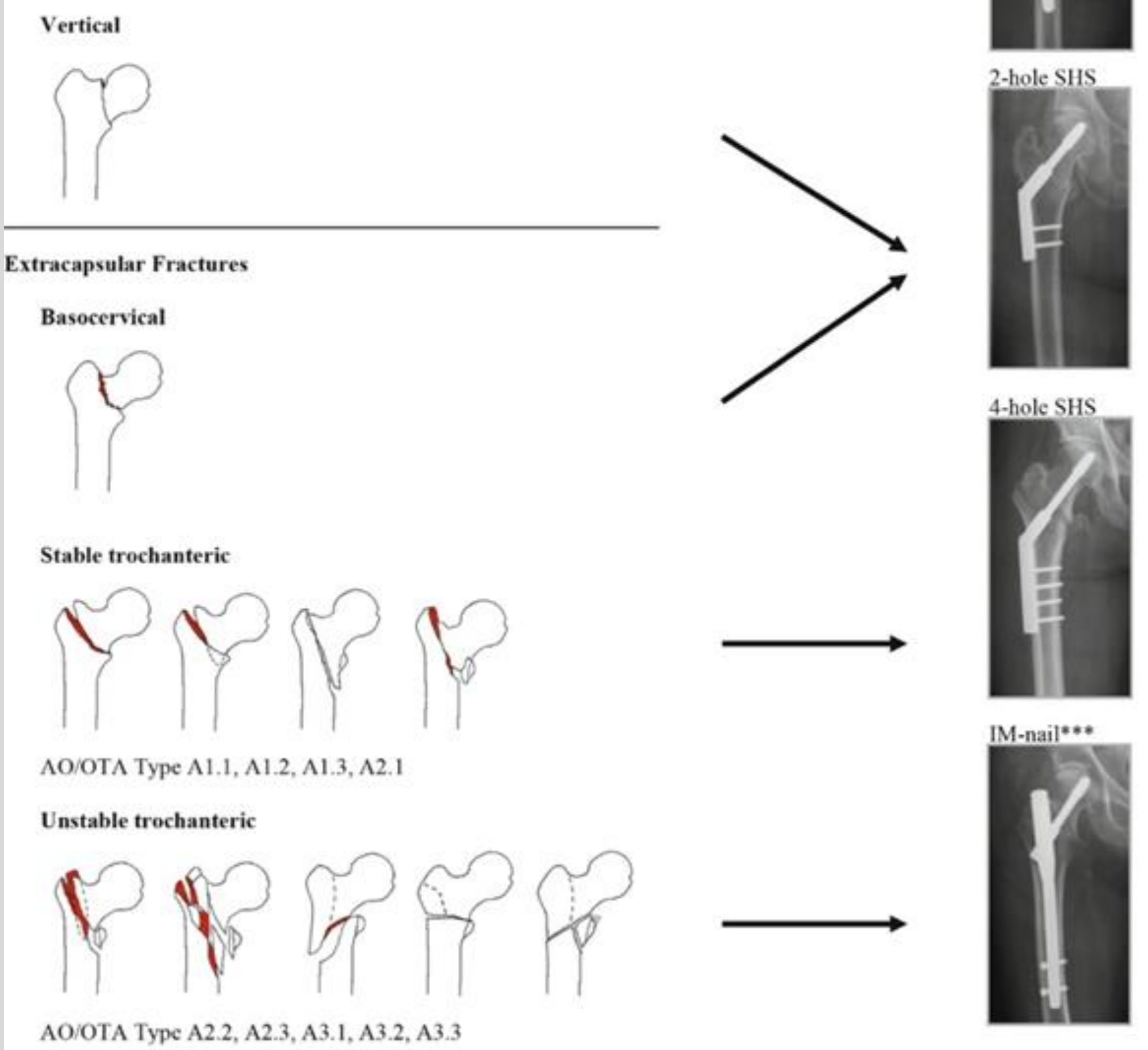


https://www.ncbi.nlm.nih.gov/books/NBK543818/figure/ch5.Fig2/

DO NOT MEMORIZE (not tested, just interesting)



https://www.ncbi.nlm.nih.gov/books/NBK543818/figure/ch5.Fig2/

Femoral Shaft Fractures

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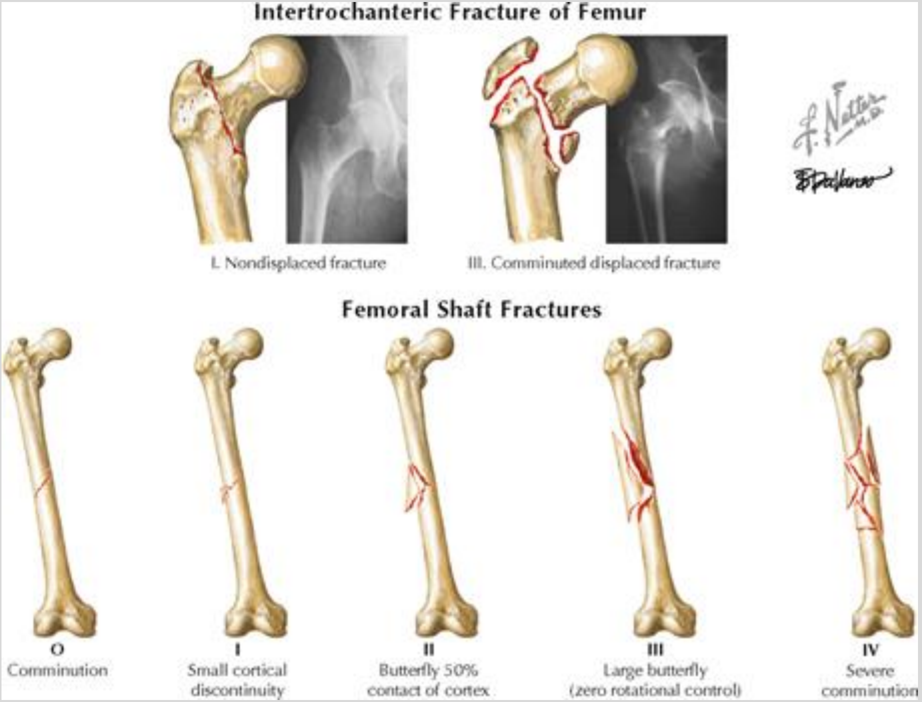
- Orthopaedic emergency
- Potential source of significant blood loss
- Monitor for Compartment Syndrome



- Thompson, Jon C. Netter's Concise Orthopedic Anatomy, 2nd Edition. 2010

Femur Fracture
-> Treat with Intramedullary Nail

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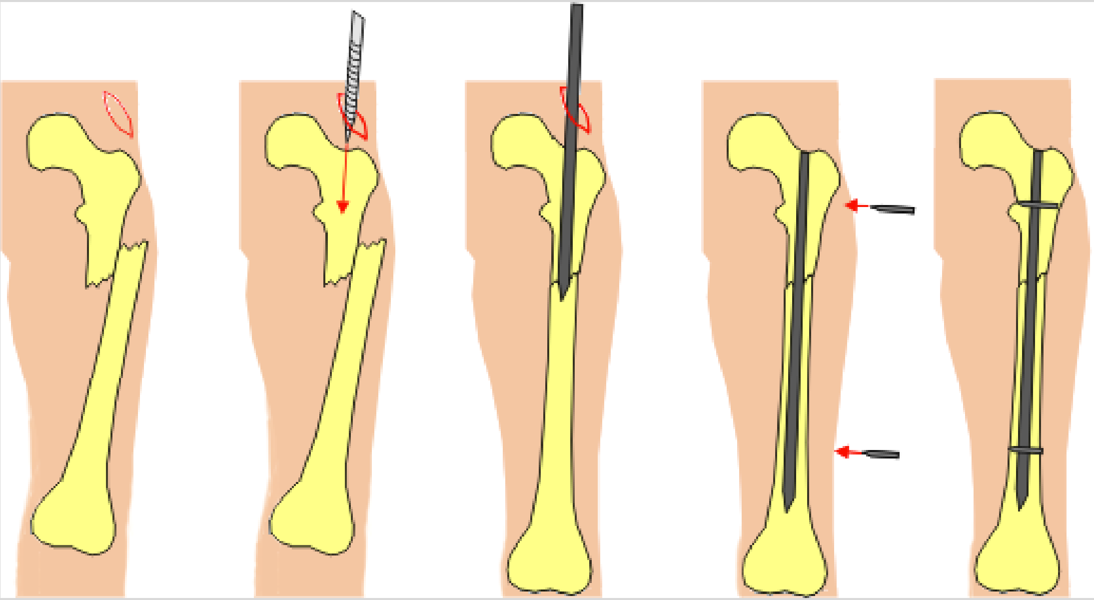

https://orthoinfo.aaos.org/en/diseases--conditions/femur-shaft-fractures-broken-thighbone/

https://boneandspine.com/intramedullary-nailing/

Degenerative Diseases of the Hip and Knee (e.g. Osteoarthritis)

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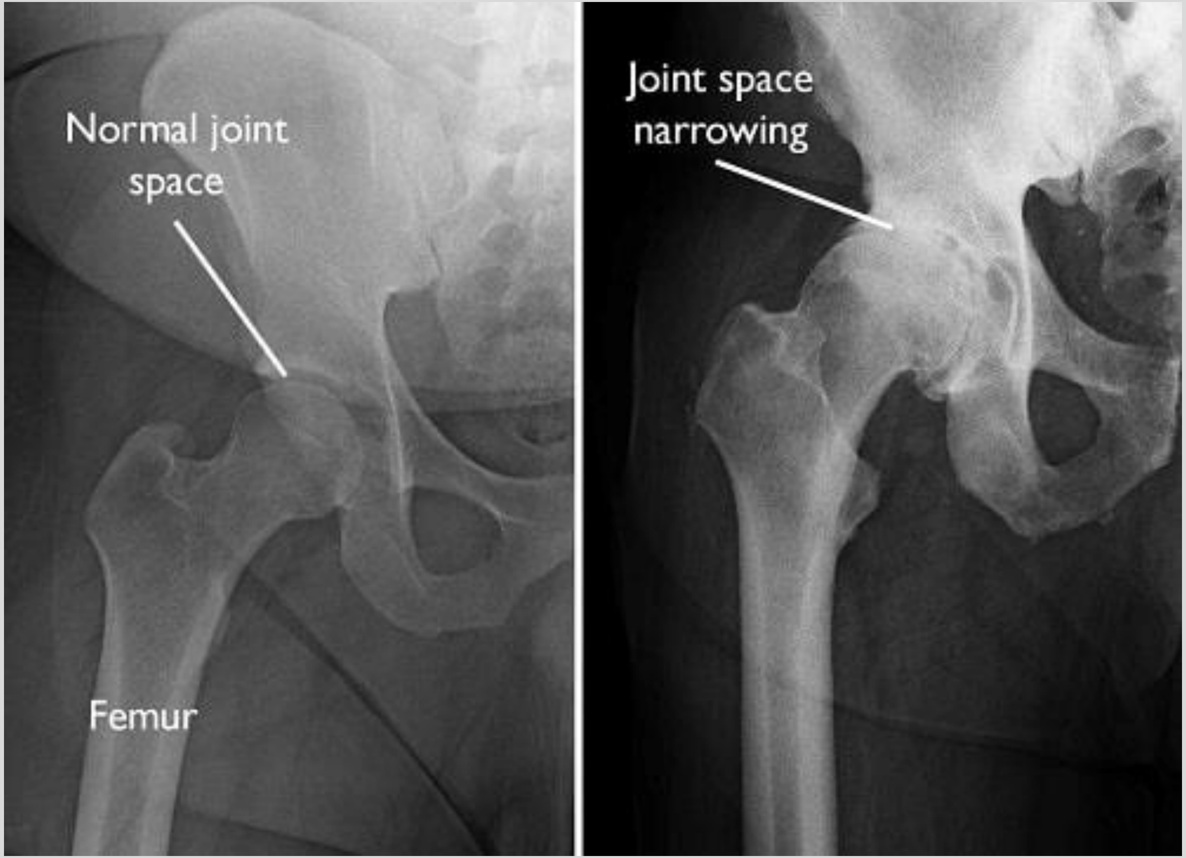
- Pain
- Limited Range of Motion
- Functional Deficits:
  - Walking, Sitting, Standing, Dressing, Personal Hygiene, Exercise, Work

- Diagnosis: X-rays

Treatment:

- RICE, activity modification, weight loss, Physical Therapy
- NSAIDS
- Pain management? (with all of its issues)
- Joint Injections
- Surgery



https://orthoinfo.aaos.org/en/diseases--conditions/osteoarthritis-of-the-hip

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Joint Injections

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- Local anesthetics
    1. Corticosteroid preparations
    2. Visco-supplementation therapy (Knee only)

        Hyaluronic acid preparations in a gel

- May give pain relief and improve quality of life, short term.
- Prolong need for surgical replacement

DOES NOT REVERSE ARTHRITIC PROCESS!

- Actually, it has been found to accelerate the progression of the Osteoarthritis

Intra-articular Corticosteroid Injections in the Hip and Knee: Perhaps Not as Safe as We Thought?

https://pubmed.ncbi.nlm.nih.gov/31617798/

Total Hip Replacement

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https://orthoinfo.aaos.org/en/treatment/total-hip-replacement/

https://www.rheumatologynetwork.com/view/total-hip-arthroplasty-no-better-partial-joint-replacement

Total Knee Replacement

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https://www.thecenteroregon.com/pain-injury-services/total-knee-replacement/

Knee

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A. Anatomy
B. Patella fractures
C. Patellar instability & dislocation
D. Collateral ligament injuries
E. Meniscal injuries
F. Cruciate ligament injuries
G. Bursitis of the knee
H. Baker’s cyst

Source: https://www.painspot.com/public/bodytool/images/female/knee/small/Knee_Front.png

Knee - Anatomy

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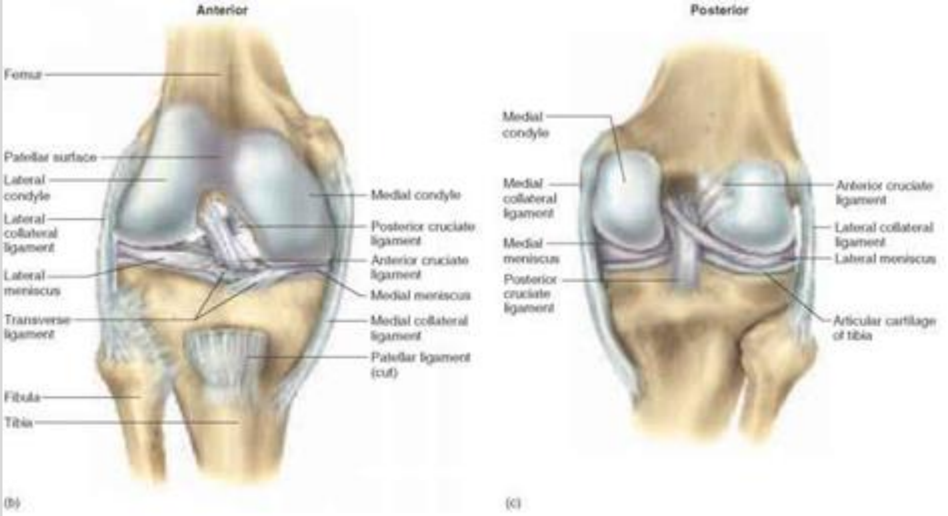
- Medial Collateral Ligament (MCL)
  - Restricts against valgus forces.
- Lateral Collateral Ligament (LCL)
  - Restricts against varus forces.
- Anterior Cruciate Ligament (ACL)
  - Restricts anterior translation & rotation of the tibia on the femur.
- Posterior Cruciate Ligament (PCL)
  - Restricts posterior translation of the tibia on the femur.
- Menisci (medial & lateral)
  - Wedge shaped fibrocartilage on the articular surfaces of the tibia
  - Shock absorption for load bearing on knee.
  - Deepen tibial surface -> aids in knee stability



Moore, K. L., Dalley, A. F.,&Agur, A. M. (2014). Lower Limb. In *Clinicall Oriented Anatomy*(7th ed., p. 636-642). Lipincott Williams and Wilkins.

Patella Fractures

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- Causes:
  - Direct impact: Falling directly onto the knee or receiving a sharp blow to the knee (MVA where knee hits dashboard directly).
  - Indirect: sudden eccentric contraction of the quads, which can pull apart the patella.
- Symptoms- bruising, inability to straighten knee, inability to walk
  - loss of connection b/w quadriceps-patellar ligament-tibia
- Imaging- Radiographs
- Treatment-
  - Immobilize knee in extension Cast/Brace + full weight bearing
  - Surgical Treatment (ORIF, patellectomy) if severe
- Complications - Popliteal artery rupture (from posterior directed force, need to assess in knee dislocations)

Source: Course https://orthoinfo.aaos.org/en/diseases--conditions/patellar-kneecap-fractures/
https://www.orthobullets.com/trauma/1042/patella-fracture

Patellar Instability & Dislocation

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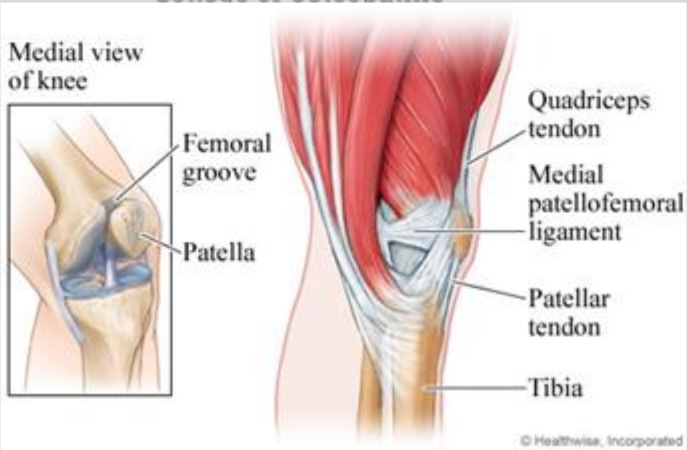
- Can occur lateral (most common), medial, vertical
- Causes:
  - If the patellar groove (aka trochlea; where the patella articulates with the femur) is too shallow or uneven, the patella can pop out of place. High energy trauma such as falls, MVAs, and sports related contact.
  - Ligamentous laxity (Ehlers-Danlos Syndrome), previous instability event
- Symptoms: Knee instability/buckling, severe ant. knee pain, increased passive patellar translation, + patellar apprehension, obvious physical deformity.
- Associations: Rupture of the medial patella-femoral ligament (MPFL), weakness of the vastus medialis.
  - these structures normally stabilize the patella
- Treatment-
  - Reduction (applying a gentle force to place patella back in place) + Conservative tx (NSAIDs, activity mod, PT)
  - MPFL repair or reconstruction



https://orthoinfo.aaos.org/en/diseases--conditions/unstable-kneecap/
http://www.myorthodoc.com/blog/patella-dislocations-and-treatment/
https://www.orthobullets.com/knee-and-sports/3020/patellar-instability
https://onsmd.com/treatment/medial-patellofemoral-ligament-mpfl-reconstruction/

Patellar Instability & Dislocation

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Patrick Mahomes’ Patella dislocation on TNF October 17, 2019:

1. https://www.youtube.com/watch?v=rxvcLVT-uRk
2. https://www.youtube.com/watch?v=8tnIGmqpYWI

https://www.myorthodoc.com/patella-dislocations-and-treatment/

Collateral Ligament Injuries

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Medial Collateral Ligament

- Causes- Direct, high energy force to the lateral side of knee (vaLgus force). Tear of MCL commonly occur w/ medial meniscus tear since their fibers are connected.
- Symptoms- Pain and swelling on the medial side of the knee. Knee instability and buckling also present.
- Examination- +Apley distraction, Increased laxity with valgus stress at 0 and 30 degrees of flexion.
- Treatment- Hinged knee brace; surgical repair (for complete tear)

Lateral Collateral Ligament (rarely injured in isolation)

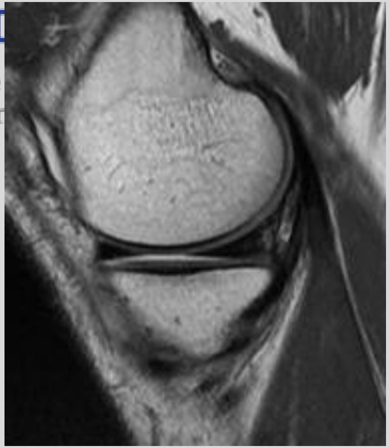
- Causes- Direct, high energy force to the medial side of knee (varus force).
- Symptoms- Pain and swelling on the lateral side of the knee. Knee instability and buckling also present.
- Examination- +Apley distraction, Increased laxity with varus stress at 0 and 30 degrees of flexion.
- Treatment- Hinged knee brace; surgical repair (for complete tear)

Source: 1. http://www.fpnotebook.com/mobile/Ortho/Exam/KnVlgsStrsTst.htm

2. :Moore, K. L., Dalley, A. F.,&Agur, A. M. (2014). Lower Limb. In *Clinically Oriented Anatomy*(7th ed., p. 662). Lipincott Williams and Wilkins.

3. https://www.orthobullets.com/knee-and-sports/3010/mcl-knee-injuries

Meniscal Tears



- 2 Causes- traumatic and degenerative
  - Traumatic Injury- Most often occurs in sports players who either rapidly twist their knee or experience a direct contact such as a tackle
  - Degenerative Injury- Occurs most often in elderly people whose menisci underwent age-related degeneration. Any sudden twisting movement, such as getting up from a chair, can result in a tear.
- Medial meniscus is more commonly torn than lateral meniscus
  - the exception is in the setting of an acute ACL tear where lateral tears are more common
- Symptoms- Pain, stiffness, delayed swelling, locking, decreased ROM, "popping" noise when injury first occurs.
- Examination- +McMurray Test, + Apley compression, joint line tenderness
- Treatment-
  - Degenerative: Rest, Ice, PT
  - Traumatic (young pts): meniscal repair, partial/total meniscectomy +/- transplant
- Location of injury affects healing
  - Peripheral 1/3 (red zone; vascularized): best healing potential
  - Inner 2/3 (white zone; avascular): no intrinsic healing ability

Moore, K. L., Dalley, A. F.,&Agur, A. M. (2014). Lower Limb. In *Clinically Oriented Anatomy*(7th ed., p. 663). Lipincott Williams and Wilkins.
https://www.orthobullets.com/knee-and-sports/3005/meniscal-injury
https://orthoinfo.aaos.org/en/diseases--conditions/meniscus-tears/
http://www.fpnotebook.com/mobile/Ortho/Exam/McmryTst.htm
https://www.google.com/url?sa=i&source=images&cd=&cad=rja&uact=8&ved=2ahUKEwiptsqoilzgAhXnm-AKHXpeBaIQjRx6BAgBEAU&url=http%3A%2F%2Fradsource.us%2Fmeniscal-tear-patterns%2F&psig=AOvVaw1IM16_1h4To-Juo7ArqLOe&ust=1548613249661579

Anterior Cruciate Ligament (ACL) Tear

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- Common, comprising 40-50% of all ligamentous knee injuries
- Female athletes 2-8x higher risk of ACL tears
  - Wider hips -> ↑Q-angle -> Greater valgus knee loading in landing
- Cause- Hyperextension and severe force directed anteriorly with the knee semiflexed, causing tibia to slide anteriorly against the femur. "Popping" noise is often heard when the injury first occurs.
  - Non-contact pivot/changing direction (Most common), stopping suddenly, direct blow, poor landing from a jump often cause ACL rupture.
  - Often a NONCONTACT athletic injury
- Symptoms-Pain, swelling, decreased ROM, discomfort while walking. Hemarthrosis (bleeding into joint space) is often found on aspiration.
- Examination- + Lachman test (most sensitive), + anterior drawer test, X-Ray to rule out any associated fractures, MRI for confirmation
- Treatment:
  - Low demand pt (e.g. elderly): PT + lifestyle mods
  - Active pt: Surgical reconstruction

Source: 1. Miller's Review of Orthopaedics: Seventh Edition. 349-352.
2. Netter's Orthopaedics. First Edition: 419-420
3. Moore, K. L., Dalley, A. F.,&Agur, A. M. (2014). Lower Limb. In *Clinically Oriented Anatomy*(7th ed., p. 663). Lipincott Williams and Wilkins.

Superbowl LVI- Odell Beckham Jr. ACL injury- 2/13/2022

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Screenshots from: https://www.youtube.com/watch?v=6jIgR-5aboc

https://www.cbssports.com/nfl/news/super-bowl-2022-odell-beckham-jr-s-injury-causes-nfl-stars-to-advocate-for-banning-turf-field-in-stadiums/

O’Donoghue’s Triad: “Terrible (Unhappy) Triad”

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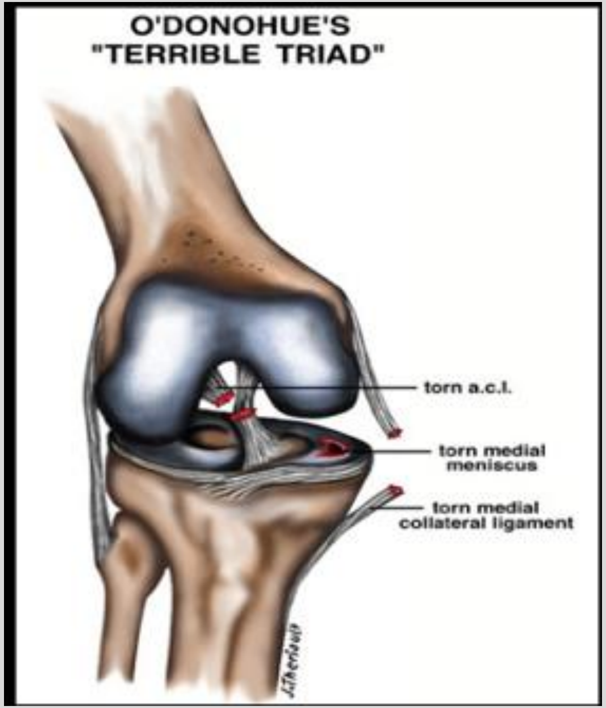
- Triad: Injury to the ACL, MCL, and medial meniscus

- This is a classic triad; however, it is more common for the lateral meniscus to be torn in acute injuries to the ACL
  - Valgus force to the knee leads to compression of the lateral meniscus between the lateral femoral condyle and lateral tibial plateau



https://classconnection.s3.amazonaws.com/490/flashcards/1762490/png/screen_shot_2012-10-15_at_54303_pm1350348195993.png
http://www.emedx.com/emedx/diagnosis_information/diagnosis_information_image_files/knee_images/acl_mri_normal.JPG
http://www.emedx.com/emedx/diagnosis_information/diagnosis_information_image_files/knee_images/acl_mri_torn.JPG

Posterior Cruciate Ligament (PCL) Tear

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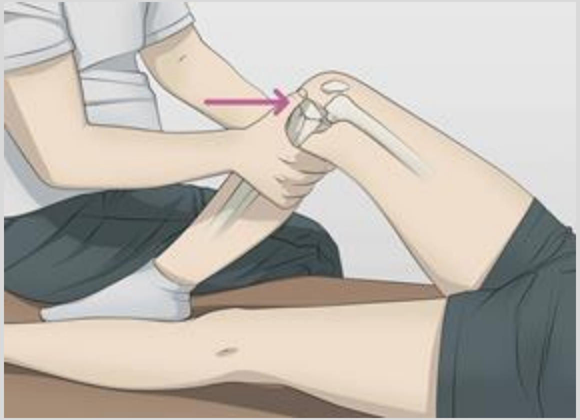
- The PCL is a strong ligament, so ruptures rarely occur.
- Causes: Contact injury (usually)
  - Fall/blow to proximal tibia w/ flexed knee (MVA dashboard injury) - tibia moves posteriorly against the femur.
  - Hyperextension injury
- Symptoms- Pain, swelling, difficulty walking, instability
- Examination- +Posterior drawer test, X-Ray to rule out concurrent bone fractures, MRI to confirm diagnosis.

Treatment- Surgery only needed when concurrent injuries (bone fractures, LCL, or MCL tears) have occurred.

Otherwise heals well with RICE (rest, ice, compression, elevation) protocol, Rehab, and bracing.



Source: 1. Moore, K. L., Dalley, A. F.,&Agur, A. M. (2014). Lower Limb. In *Clinically Oriented Anatomy*(7th ed., p. 663). Lipincott Williams and Wilkins.

https://orthoinfo.aaos.org/en/diseases--conditions/posterior-cruciate-ligament-injuries/

Posterior Cruciate Ligament (PCL) Tear

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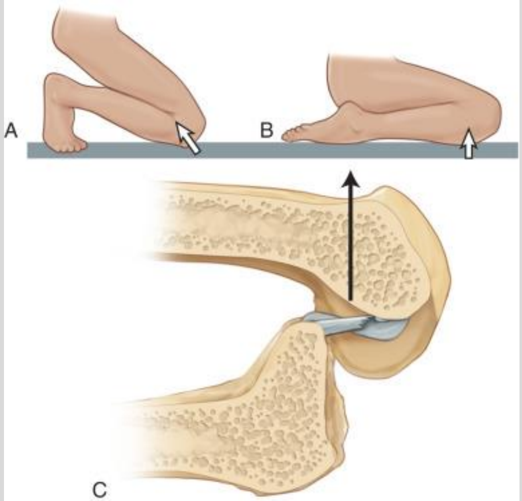

Source: DeLee, Drez, & Miller's Orthopaedic Sports Medicine

ACL vs. PCL

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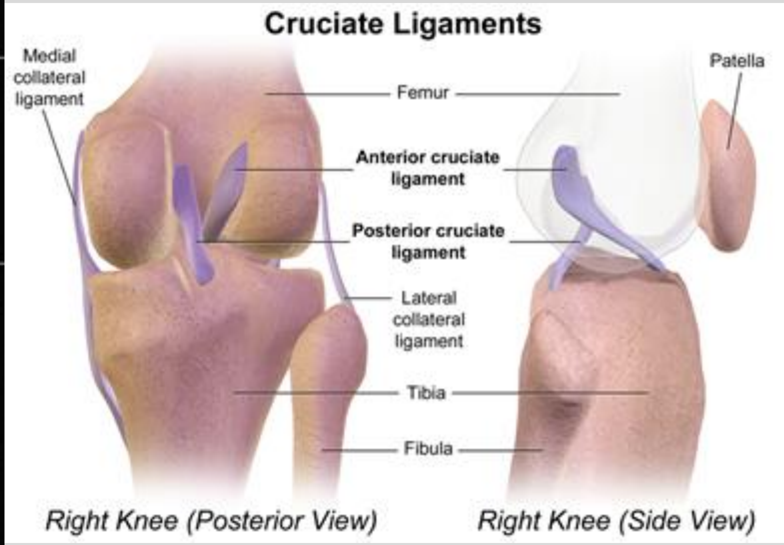

https://www.orthobullets.com/knee-and-sports/3005/meniscal-injury
https://radiopaedia.org/cases/knee-mri-sagittal-pd-1

https://www.physio-pedia.com/File:Cruciate_Ligaments.png

Knee - Bursitis

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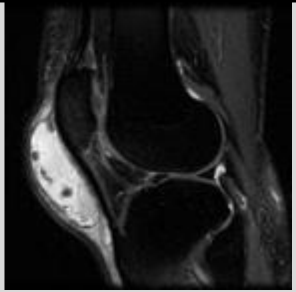
- Bursa: fluid filled sacs lined by synovial membrane that cushion and ↑gliding between bones and tendons/muscles

- Prepatellar bursitis (most common)
  - Path: Inflammation of prepatellar bursa
  - Causes:
    - Repetitive trauma/pressure to knee (e.g. kneeling, wrestling); "Housemaid's knee"
  - Symptoms
    - Pain with ACTIVE ROM (not passive), swelling anterior to patella; warmth (especially if septic)
  - Diagnosed clinically
  - Tx: NSAIDs, compressive wrap, +/- aspiration & abx



https://www.orthobullets.com/knee-and-sports/3018/prepatellar-bursitis-housemaids-knee?expandLeftMenu=true

Knee - Bursitis, Cont.

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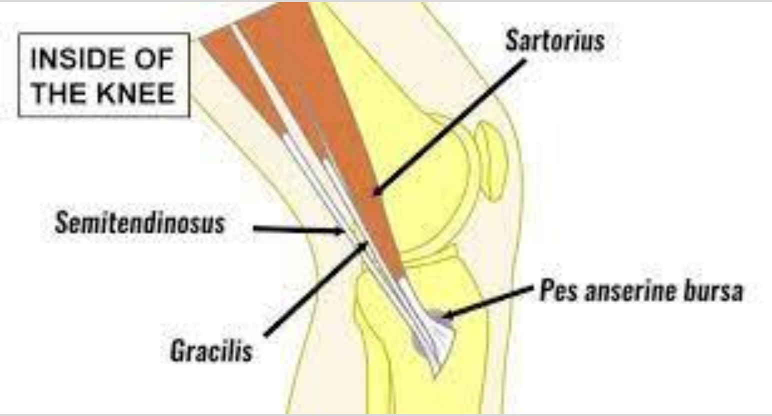
- Anserine Bursitis (Pes anserinus pain syndrome)
  - Etiology: Obesity, overuse in athletes
  - Sharp, non-radiating pain on medial side of tibia just below the knee (semitendinosus, gracilus, saritorius)
  - No pain or laxity with valgus stress test (r/o MCL)



Baker's Cyst (Popliteal Cyst)

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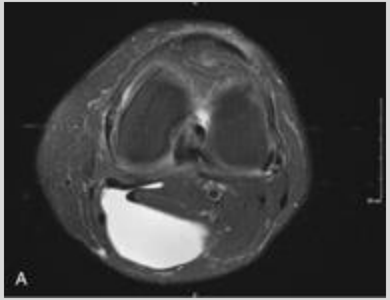
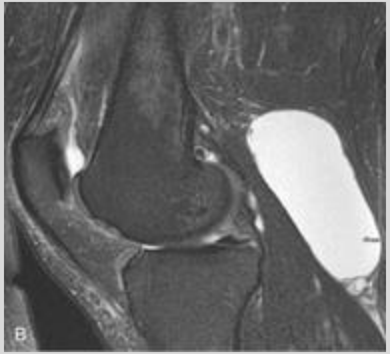
- Fluid collection in popliteal bursa behind the knee
    - Cyst forms in the bursa underlying the Gastrocnemius & Semimembranosus tendons
- This bursa communicates directly with synovial space of knee joint. Chronic joint disease of the knee (e.g. any form of arthritis) causes fluid collection in this bursa.



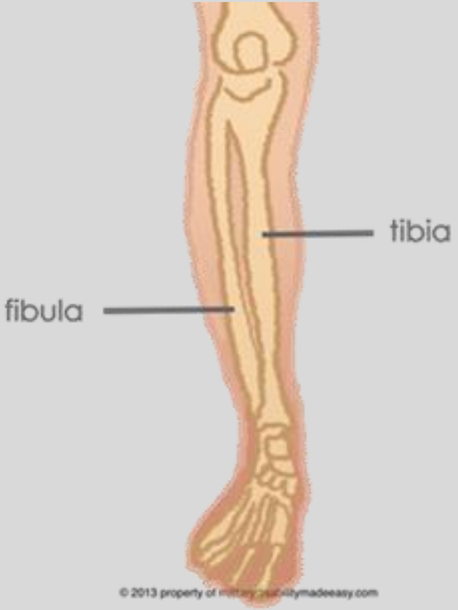


- <u>Common causes:</u>
    - Osteoarthritis
    - Rheumatoid arthritis
    - Joint injury (meniscal tears, ACL rupture)
    - Gout/pseudogout
- <u>Symptoms:</u>
    - Generally asymptomatic; May cause posterior knee pain, pain with prolonged standing or activity
    - Rupture causes acute severe pain that can mimic symptoms of DVT
- <u>Diagnostic Imaging:</u>
    - Ultrasound (r/o DVT); MRI

Miller's Essential Orthopaedics, 2010 Chapter 152, 657-659

Lower Leg Pathologies

1. Compartment syndrome
2. Tibial fractures
3. Venous thromboemobolism (VTE)



Source: http://www.militarydisabilitymadeeasy.com/kneeandleg.html

Lower Leg - Compartment Syndrome

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- Cause: severe trauma (e.g. MVA, fracture)
- Pathophysiology:
  - Swelling or bleeding w/in a compartment
  - Each compartment is lined by fascia, an un-stretchable substance -> swelling or bleeding leads to increased pressure.
  - pressure build up in one of the compartments -> prevents blood flow from reaching nerve and muscle cells.
- Anterior compartment is most commonly affected.

1. Moore, K. L., Dalley, A. F.,&Agur, A. M. (2014). Lower Limb. In *Clinically Oriented Anatomy*(7th ed., p. 608-609). Lipincott Williams and Wilkins.
2. https://orthoinfo.aaos.org/en/diseases--conditions/compartment-syndrome/

Acute Compartment Syndrome

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- Causes:
  - Fracture
  - Bruised Muscle/Hematoma
  - Reestablished blood flow after blocked circulation (ie. Reperfusion of acute limb ischemia)
  - Crush injuries
  - Casts or tight bandages (prolonged compression)
- Symptoms:
  - Pain more intense than what is expected from initial injury. Usually in the front of the leg (anterior compartment).
  - ↑ pain with passive stretch; e.g. pain with plantar flexion for anterior compartment
  - Paresthesia (early) - Tingling or burning sensations on skin based on which compartment affected. (sensory nerve ischemia)
  - Numbness or paralysis are late signs
- Diagnostic testing- Measure compartment pressures, radiographs to r/o fx
- Treatment - Surgical emergency! In all compartments, increased pressure can lead to tissue death. An incision through the fascia (fasciotomy) must be made to relieve the pressure if it is above 30 mmHg.

https://orthoinfo.aaos.org/en/diseases--conditions/compartment-syndrome/
https://www.orthobullets.com/trauma/1001/leg-compartment-syndrome

Fasciotomy for Compartment Syndrome

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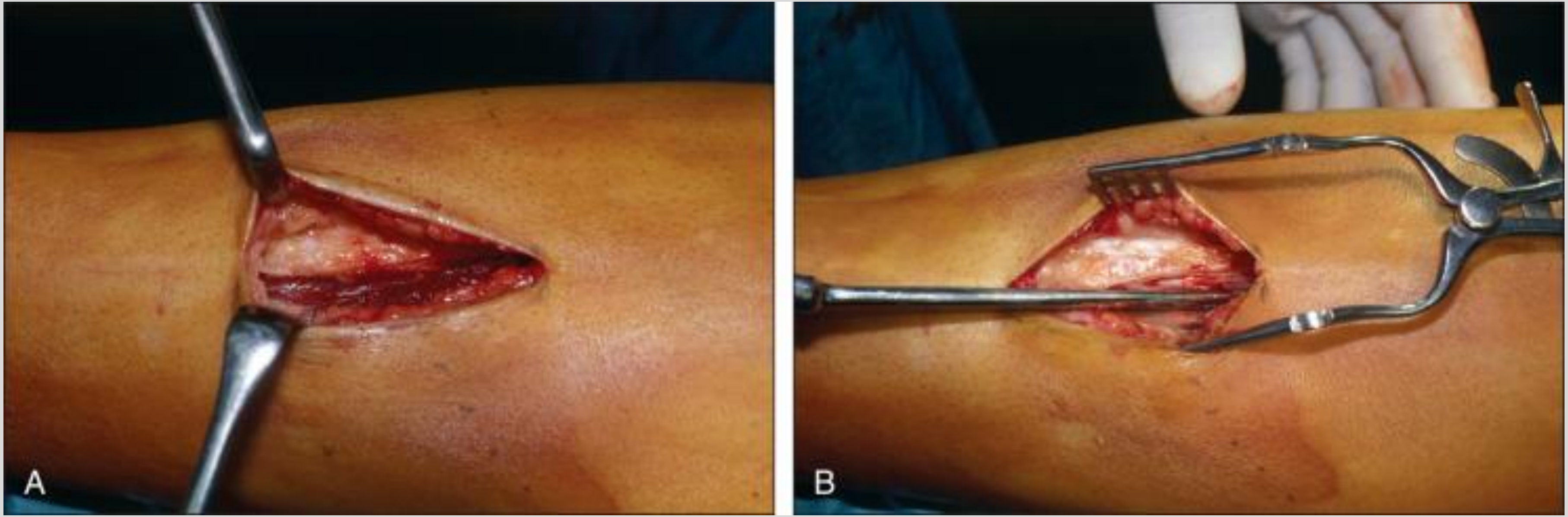

Source: DeLee, Drez, & Miller’s Orthopaedic Sports Medicine

Lower Leg Compartments

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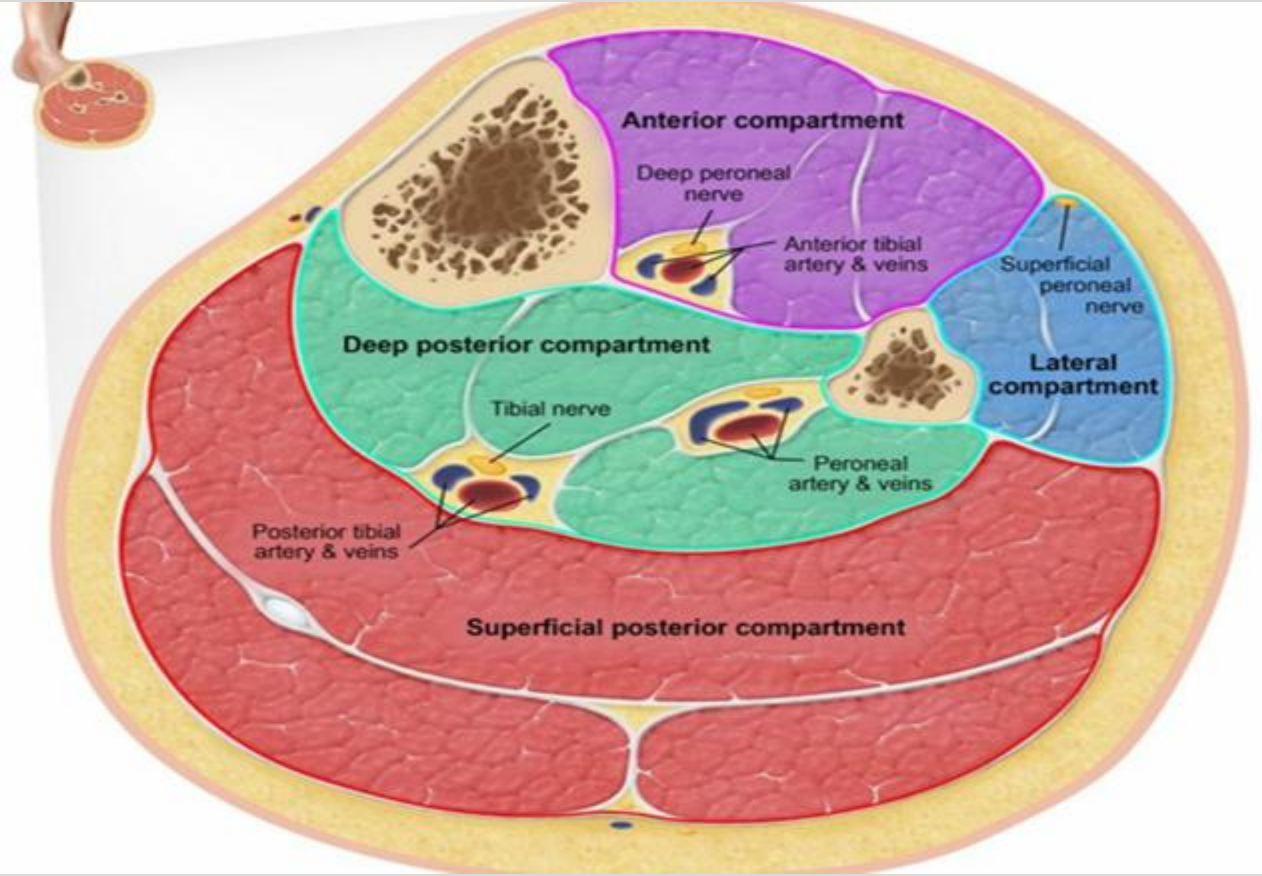
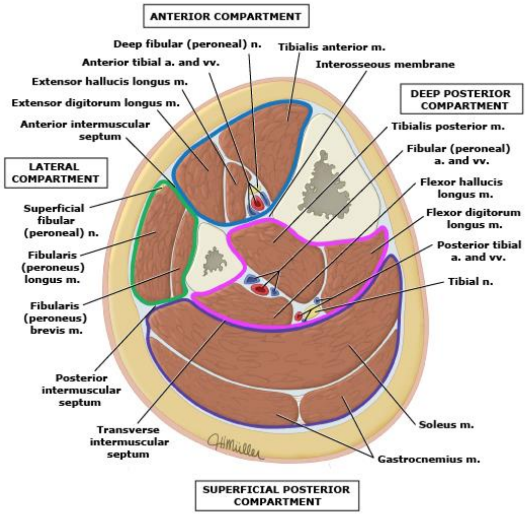


https://www.orthobullets.com/anatomy/10133/compartments-of-leg

Muscle compartments of the leg



Compartments of the leg:

- **Anterior compartment (blue):** Tibialis anterior, extensor muscles of the foot, and fibularis (peroneus) tertius muscles. The anterior tibial artery and deep fibular (peroneal) nerve supply the anterior compartment.
- **Lateral compartment (green):** Fibularis (peroneus) longus and fibularis (peroneus) brevis muscles. The superficial fibular (peroneal) nerve and branches from the anterior tibial artery supply these muscles.
- **Superficial posterior compartment (purple):** Gastrocnemius, soleus, and plantaris muscles. Tibial nerve branches supply these muscles. The arteries that supply these muscles descend from the popliteal artery. The sural arteries (medial, lateral) supply the gastrocnemius. The soleus is variably supplied by the popliteal artery, posterior tibial artery, and fibular (peroneal) artery.
- **Deep posterior compartment (pink):** Tibialis posterior, flexor muscles of the foot, and popliteus muscles. The deep posterior compartment is innervated by the tibial nerve and supplied by the posterior tibial and fibular (peroneal) arteries.

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1. Anterior

2. Lateral

3. Superficial Posterior Compartment

4. Deep Posterior Compartment

Source: Uptodate: Muscle Compartment of the leg

Lower Leg - Compartment Syndrome

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Compartment	Nerve Innervation	Pain Location	Numbness Location	Action that causes pain
Anterior	Deep Peroneal Nerve	Anterior leg	First web space	Plantar flexion
Lateral	Superficial Peroneal Nerve	Lateral leg	Dorsum of foot	Inversion and dorsiflexion
Superficial Posterior	Tibial Nerve	Posterior leg	Lateral foot	Dorsiflexion
Deep Posterior	Tibial Nerve	Posterior leg	Plantar aspect of foot	Dorsiflexion

http://boneandspine.com/compartment-syndrome/

Lower Leg - Chronic (exertional) Compartment Syndrome

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- Causes:
  - Exercise with repetitive motions such as running, biking, or swimming.
  - "stiff" fascia does not allow for the increased blood flow that exercising muscle requires
- Symptoms:
  - Achy, crampy, squeezing pain or pressure
    - Reproducible by the patient (e.g. begins within several minutes during running/training)
    - Cessation of exercise usually decreases pain.
  - Paresthesias, numbness, foot-drop
- Diagnosis- Direct measurement of intramuscular compartment pressures
- Treatment:
  - Non-surgical- Physical therapy, orthotics, NSAIDS, cessation/modification of activity.
  - Surgical- If conservative measures fail, then a fasciotomy may be needed.

https://www-uptodate-com.arktos.nyit.edu/contents/chronic-exertional-compartment-syndrome?search=chronic%20compartment%20syndrome&source=search_result&selectedTitle=1~14&usage_type=default&display_rank=1
https://orthoinfo.aaos.org/en/diseases--conditions/compartment-syndrome/

Lower Leg - Tibial Fractures

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- Cause- High velocity impact to the tibia
- Symptoms:
  - Pain
  - Inability to bear weight
  - Leg deformity
  - Bone protruding through the skin
  - Loss of sensation in foot- rare
- Diagnostic Tests:
  - X-Ray
  - CT
- Treatment- Depends on the type & severity of fracture.
  - Non-surgical candidates will first receive a splint until the swelling goes down and then either a cast or brace.

Source: Schaller, T. Essential Orthopedics 2020.
Left: Plate Fixation
Right: Intramedullary implant

https://orthoinfo.aaos.org/en/diseases--conditions/tibia-shinbone-shaft-fractures/

Tibial Fractures - Alex Smith

https://www.espn.com/nfl/story/_/id/29112995/alex-smith-comeback-fight-save-qb-leg-life

Tibial Fractures - Kevin Ware

https://medium.com/@averyp729/kevin-ware-is-he-now-the-story-of-the-most-gruesome-and-disgusting-injury-in-basketball-history-8aea56cd6c95

Venous thromboembolism (VTE)/Deep Venous Thrombosis (DVT)

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- A venous thrombus is a blood clot that forms within a vein.
- Most common locations are the deep veins of the legs
  - Femoral, popliteal veins
- Virchow's Triad- 3 causes that lead to thrombus formation
  - Stasis of blood flow
  - Endothelial Damage
  - Hypercoagulable state
- Common Risk factors
  - Old age, Trauma (endothelial damage + immobilization), Surgery (Pts should receive DVT prophylaxis & monitoring), Malignancy (hypercoagulable state), Pregnancy (hypercoagulable state), Inherited thrombophilia (hypercoagulable state), Nephrotic syndrome (hypercoagulable state), Long distance travel, Med-induced hypercoagulable state (OCPS, testosterone, hormone replacement therapy, SERMs)
- DVT can embolize to lungs (PE) or to the brain/systemic circulation (if there is a PFO/ASD)

Miller's Essential Orthopaedics, 2010 Chapter 20, 92-94
https://www.orthobullets.com/basic-science/9056/thromboembolism-pe-and-dvt?expandLeftMenu=true

DVT/VTE

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- Symptoms:
  - U/L leg pain (eg Calf pain)
  - Palpable Cord felt on back of leg (from a thrombosed vein)
  - Unilateral edema (poor venous outflow from leg)
  - Warmth, tenderness, erythema in calf or thigh
  - + Homan's Sign: increased calf pain with ankle dorsiflexion
- Diagnostic Tests:
  - Use Wells' Criteria to aid in diagnosis of DVT/PE
  - Lower extremity Compression ultrasound w/ doppler flow studies (aka Duplex Ultrasonography)
    - Initial diagnostic test of choice -> look for non-compressibility of affected vein
  - D-dimer level: fibrin breakdown product -> ↑clotting -> ↑clot breakdown -> ↑D-dimer
- Treatment:
  - Anticoagulation with unfractionated heparin or LMWH (Transition to Warfarin for LT prevention)
- Prophylaxis:
  - Early fracture stabilization & mobilization
  - Mechanical venous circulatory assist (calf pumps/SCD/IPC)
  - Pharmacologic prophylaxis (anti-coagulants)

Miller's Essential Orthopaedics, 2010 Chapter 20, 92-94

Foot and Ankle

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A. Low ankle sprains
B. High ankle sprains
C. Ottawa ankle rules
D. Achilles tendon rupture

Medial and Lateral Ankle Ligaments

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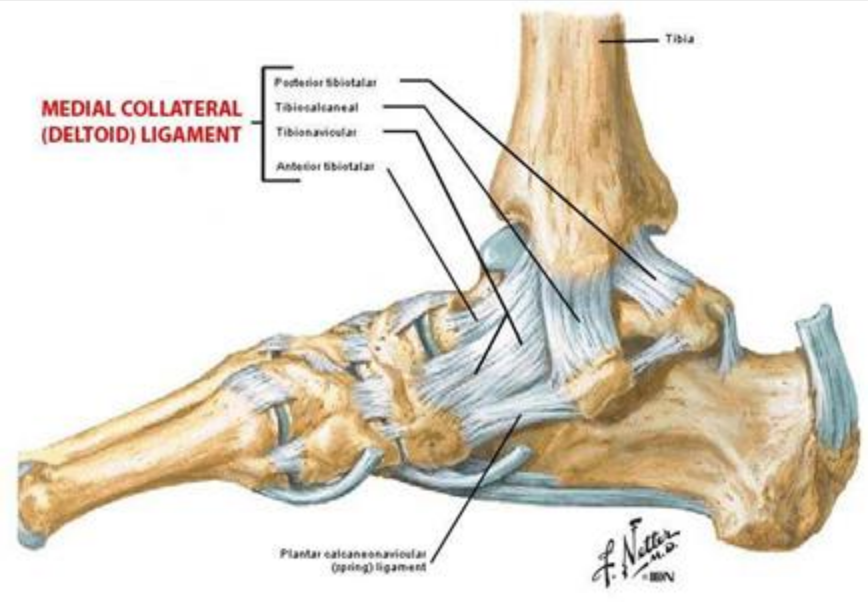
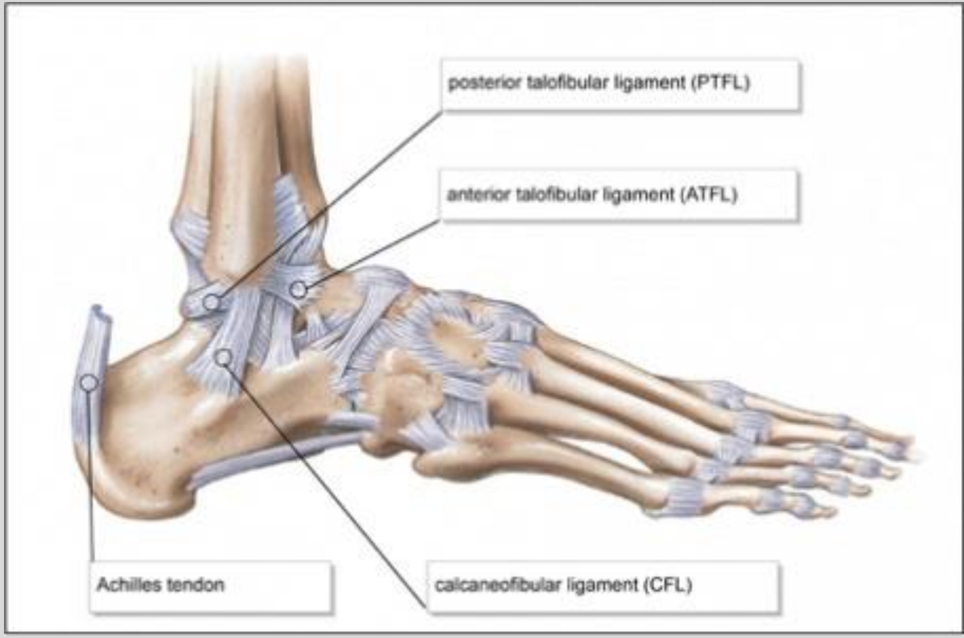
Medial Ankle Ligaments



Lateral Ankle Ligaments



Deltoid ligament - main role is to resist EVERSION

Lateral ligaments - main role is to resist INVERSION

https://www.physio-pedia.com/Stress_tests_for_Ankle_ligaments

Classification of Ankle Sprains

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- Grade I- Partial tear of a ligament with no mechanical instability
- Grade II- Incomplete tear of a ligament with moderate functional impairment
- Grade III- Complete tear of a ligament with moderate to severe functional impairment

Low Ankle Sprain - Inversion Injuries

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- Causes: Happen when the foot rolls inward, stressing the lateral ligaments of the ankle.
- Presentation: Possible audible "pop" followed by swelling, ecchymosis, and pain with weight bearing.
- Examination: Palpation, anterior drawer test of the ankle, talar tilt test, and possible X-Ray depending on Ottawa ankle rules.
- Treatment: RICE, bracing, physical therapy, possible surgery depending on grade and severity of sprain.
- Anterior Talofibular Ligament is most vulnerable
- Mnemonic: ATF- Always Tears First

https://www.med.unc.edu/fammed/fammedcenter/about-us/services/sportsmedicine/Ankle_Strengthening_handout.pdf
https://foxvalleyimaging.com/what-you-need-to-know-about-a-sprained-ankle/

Low Ankle Sprain - Eversion Injuries

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- Causes: Happen when the foot rolls outward, stressing the medial ligaments of the ankle (deltoid ligament).
- Presentation: Immediate pain and swelling with possible ecchymosis.
- Examination: Palpation, X-Ray depending on Ottawa ankle rules.
- Treatment: RICE, bracing, physical therapy, possible surgery depending on grade and severity of sprain.
- Deltoid ligament affected.
- Recovery can take twice as long as lateral ankle sprains.

http://www.sportsinjuryclinic.net/sport-injuries/ankle-achilles-

High Ankle Sprain

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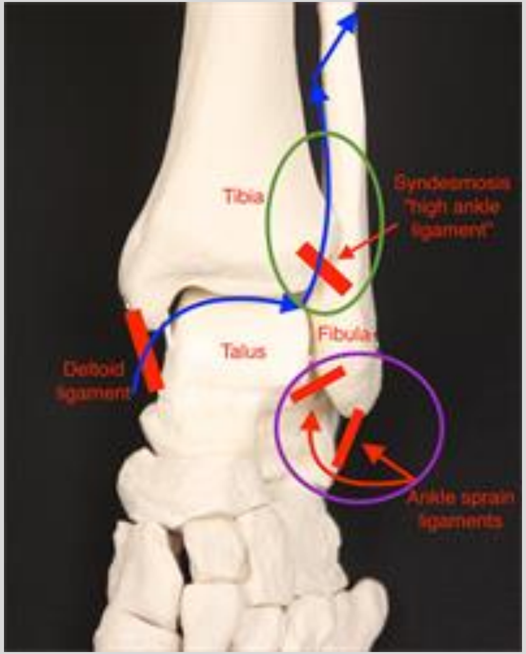
- Sprains involving the distal tibiofibular syndesmosis = Anterior inferior tibiofibular ligament (AITFL), require nearly twice as long to return to activity than "low" ankle sprains
  - 1-10% of all ankle sprains
- <u>Mechanism</u>: forced *dorsiflexion* and external rotation of ankle
- <u>Presentation</u>: Pain at the anterior syndesmosis, swelling may be minor or nonexistent
- <u>Examination</u>: squeeze test elicits pain with compression, external rotation test, x-rays



http://www.aofas.org/footcaremd/conditions/ailments-of-the-ankle/Pages/High-Ankle-Sprain.aspx

High Ankle Sprain

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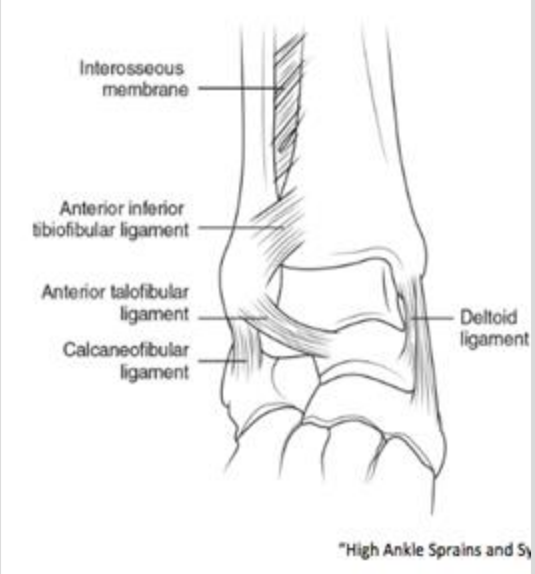

https://www.nuemblog.com/blog/high-ankle-sprain

https://www.nuemblog.com/blo
g/high-ankle-sprain

High Ankle Sprain

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https://www.youtube.com/watch?v=k2MGfCfbF64

High Ankle Sprain

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https://www.youtube.com/watch?v=k2MGfCfbF64

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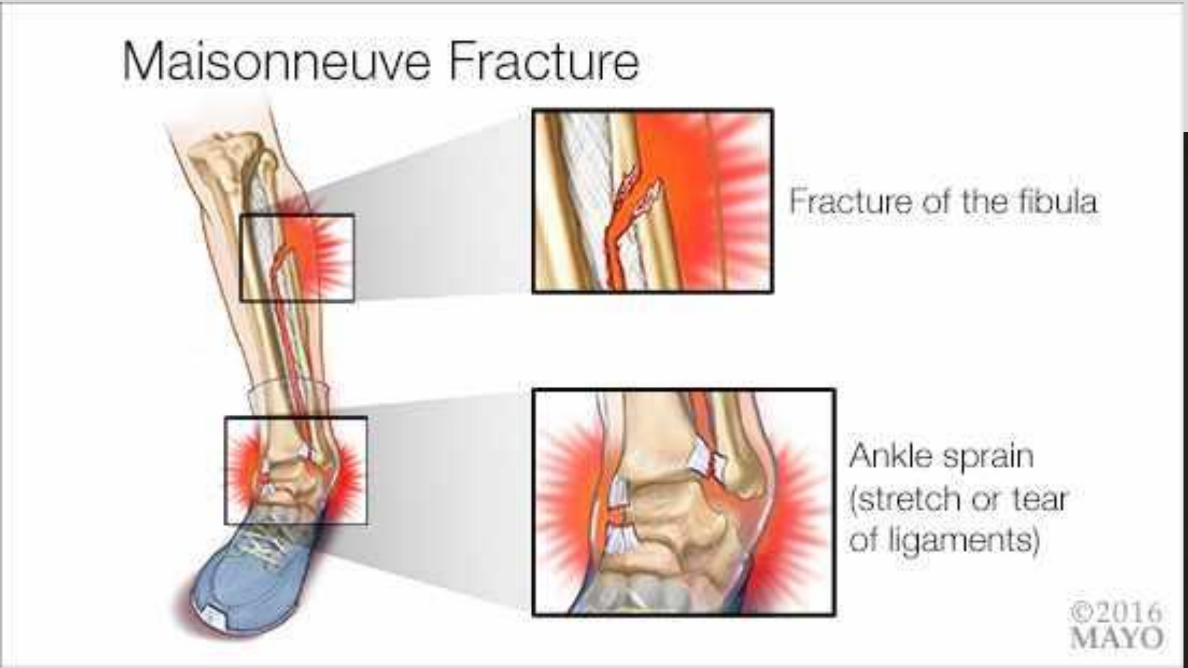

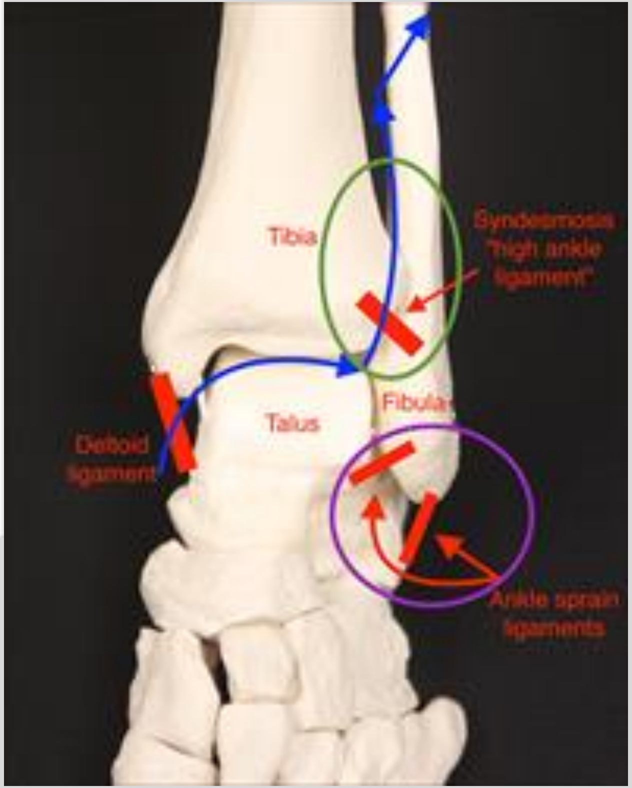

https://newsnetwork.mayoclinic.org/discussion/mayo-clinic-q-and-a-untreated-maisonneuve-fracture/

http://www.aofas.org/footcaremd/conditions/ailments-of-the-ankle/Pages/High-Ankle-Sprain.aspx

Ottawa Ankle Rules

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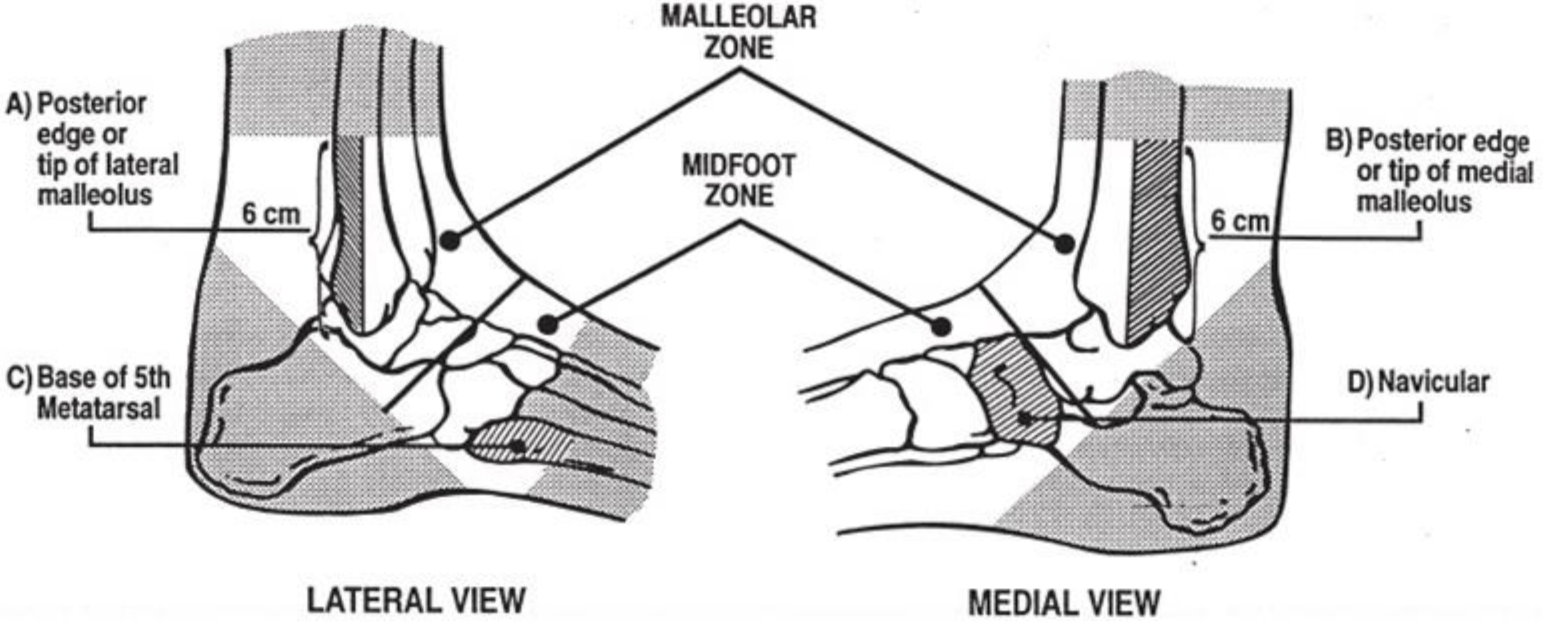
- Rules that dictate whether a patient presenting with ankle or foot pain is in need of a radiograph.
- Mnemonic: In Ottawa we drink "**Fine Malbec**"
- For Foot: FI-Ne
  - Point tenderness at the base of the FIfth metatarsal.
  - Point tenderness at the Navicular.
  - Inability to BEar weight (four steps) immediately and in emergency department.
- For Ankle: MALBEc
  - Point tenderness at posterior edge (of distal 6 cm) of lateral MALleolus.
  - Point tenderness at posterior edge (of distal 6 cm) of medial MALleolus.
  - Inability to BEar weight (four steps) immediately and in emergency department.

https://radiopaedia.org/articles/ottawa-ankle-rules-1

Ottawa
Ankle Rules

For Ankle Injury Radiography



Stiell IG, McKnight RD, Greenberg GH, McDowell I, Nair RC, Wells GA, Johns C, Worthington JR. Implementation of the Ottawa ankle rules. JAMA. 1994 Mar 16;271(11):827-32.

Achilles Tendon Rupture

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- Causes: Frequently seen in athletes, particularly males in 4th decade. Usually in zone about 6 cm above the calcaneal insertion (Hypovascular region of the tendon)
- Presentation: Usually occurs during push-off during running or a hard lateral movement. May hear "POP" as a sensation of getting hit in the back. Unable to push off with the ankle. Localized posterior ankle pain
- Examination: Loss of resistance to passive dorsiflexion or loss of active plantar flexion. Palpable gap. Bruising. Thompson Test.
- Treatment: Non-surgical→ Begins with immobilization in plantarflexion, with progressive dorsiflexion as healing progresses. Early functional rehabilitation

  Surgical→ Repairing the ruptured ends of the tendon suing sutures

- Kraeutler, Matthew J. Netter's Sports Medicine. 3rd Edition. 2023.

Achilles Tendon Rupture

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Screenshot from: https://www.youtube.com/watch?v=ruYx3gevWoE

Screenshot from: https://www.youtube.com/watch?v=ruYx3gevWoE

https://www.dailymail.co.uk/news/article-2308532/Kobe-Bryant-likely-season-snapping-Achilles-tendon-stirring-fears-mean-end-career.html

Case 1

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A 65 y/o female with hx of a.fib, PVD, and CAD comes into the ED after 12 hrs of pain and tingling in her right foot. On PE, the R. lower leg and foot are pale and cold with absent pulses however the left side has no abnormal findings. Acute limb ischemia is diagnosed and emergency surgery is performed to restore blood flow to the leg. While in recovery, she complains of intense burning pain along the lateral right leg and ankle. PE shows a normal appearing leg.

What is the most likely diagnosis?

a) Complex regional pain syndrome I

b) Complex regional pain syndrome II

c) Acute Compartment syndrome

d) Chronic Compartment syndrome

e) DVT

Answer: c) Acute Compartment syndrome

Case 1

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Which compartment is likely affected?

a) Anterior

b) Lateral

c) Superficial

d) Deep Posterior

c) Lateral

Case 1

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Which action will lead to increased pain?

a) Plantarflexion
b) Dorsiflexion
c) Inversion & dorsiflexion
d) Eversion & dorsiflexion

Answer: C) Inversion & dorsiflexion

Case 1

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Which nerve is most likely affected?

a) Deep fibular nerve
b) Superficial fibular nerve
c) Tibial Nerve

Answer B) Superficial fibular nerve

Summary Slide

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- **Know anatomy, mechanisms, symptoms, and treatments**

- Hip
  - Anatomy - blood supply. Sports injuries + degenerative joint disease.
  - Dislocations - mechanism and clinical appearance
  - Hip fractures - general mechanisms and treatment goals
  - Femur fractures - IM nail
- Knee
  - Patella fractures, instability, and dislocations
  - MCL vs. LCL, ACL vs PCL, Meniscus injuries
  - Bursitis + Baker's cyst
- Lower Leg
  - Compartment syndrome acute vs. chronic - causes, symptoms, nerves (know table on slide 57), treatments
  - Tibial fractures and DVTs
- Foot and Ankle
  - Sprains: Inversion=ATFL, eversion=deltoid ligament, high-ankle=AITFL
  - Ottawa Ankle Rules
  - Achilles tendon rupture

- Core References:

  - Moore's Clinically Oriented Anatomy, 9e, Ch. 7
  - Miller, Mark D., Orthopaedic Sports Medicine. 5th Edition. Elsevier, Inc. 2020.
  - Cleland, Joshua A. Netter's Orthopaedic Clinical Examination. 4th Edition. 2022.
  - Madden, Christopher C. Netter's Sports Medicine. 3rd Edition. 2023.

Lecture Feedback Form

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https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R

Thank you!

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If you have any questions, please reach out!

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